

Linear Algebra

1 Vector spaces

Throughout this course, $\mathbb{F}$ is a field.

Definition 1.1 – Vector space

A $\mathbb{F}$ -vector space, or a vector space over $\mathbb{F}$, is an abelian group $(V, +, \mathbf{0})$ equipped with a scalar multiplication function

$$\begin{aligned}\mathbb{F} \times V &\rightarrow V \\ (\lambda, v) &\mapsto \lambda v\end{aligned}$$

such that for all $v, w \in V$ and $\lambda, \mu \in \mathbb{F}$, we have

- (i) $(\lambda + \mu)v = \lambda v + \mu v$;
- (ii) $\lambda(v + w) = \lambda v + \lambda w$;
- (iii) $\lambda(\mu v) = (\lambda\mu)v$;
- (iv) $1v = v$.

Example 1.1 – Vector spaces

- (i) $\mathbb{F}^n$, n -dimensional column vectors with entries in $\mathbb{F}$.
- (ii) $\mathbb{C}^n$, which can be both a real or complex vector space.
- (iii) $\mathcal{M}_{m \times n}(\mathbb{F})$, matrices over $\mathbb{F}$ with m rows and n columns.
- (iv) $\mathbb{F}^X$, the space of functions from any nonempty set X to $\mathbb{F}$ with $(f + g)(x) = f(x) + g(x)$ and $(\lambda f)(x) = \lambda f(x)$.

Definition 1.2 – Vector subspace

Let V be a $\mathbb{F}$ -vector space, a subspace of V is a subset $U \subseteq V$ which is a $\mathbb{F}$ -vector space under the same operations as V , and we write $U \leq V$.

Equivalently, $(U, +)$ is a subgroup of $(V, +)$ and $\lambda u \in U$ for all $\lambda \in \mathbb{F}$ and $u \in U$. Note that the four axioms are automatically inherited in subspaces.

Proposition 1.1 – Subspace test

Let V be a $\mathbb{F}$ -vector space and $U \subseteq V$. U is a subspace of V if and only if

- (i) U is nonempty;
- (ii) $u + \lambda w \in U$ for all $u, w \in U$ and $\lambda \in \mathbb{F}$.

Proof. If U is a subspace, then U satisfies (i) and (ii).

Conversely, suppose $U \subseteq V$ satisfies (i) and (ii). Taking $\lambda = -1$ in (ii), for all $u, w \in U$, $u - w \in U$. Also, $U \neq \emptyset$. By the subgroup test from IA Groups, $(U, +)$ is a subgroup of $(V, +)$. Finally, take $u = \underline{0}$ in (ii), then for all $w \in U$ and $\lambda \in \mathbb{F}$, $\lambda w \in U$. $\square$

Example 1.2 – Vector subspaces

- (i) $\{(x \ y \ z)^\top \in \mathbb{R}^3 : x + y + z = t\}$ for fixed $t \in \mathbb{R}$ is a subspace of $\mathbb{R}^3$ if and only if $t = 0$.
- (ii) polynomial functions $P(\mathbb{R}) \leq$ infinitely differentiable functions $C^\infty(\mathbb{R}) \leq$ continuous functions $C(\mathbb{R}) \leq$ functions from the reals to the reals $\mathbb{R}^\mathbb{R}$.

Lemma 1.1

If $U, W \leq V$, then $U \cap W \leq V$.

Proof. We use the subspace test.

- (i) $U \leq V$ so $\underline{0} \in U$ and $W \leq V$ so $\underline{0} \in W$. Hence, $\underline{0} \in U \cap W \neq \emptyset$.
- (ii) For all $x, y \in U \cap W$ and $\lambda \in \mathbb{F}$, then $x, y \in U$ so $x + \lambda y \in U$. Similarly, $x + \lambda y \in W$. So, $x + \lambda y \in U \cap W$.

$\square$

The union of subspaces is almost never a subspace. We can, however, take a subspace sum.

Definition 1.3 – Subspace sum

Let $U, W \leq V$, the subspace sum of U and W is

$$U + W = \{u + w : u \in U, w \in W\}.$$

By the subspace test, if $U, W \leq V$, then $U + W \leq V$. Also, $U + W$ is the smallest subspace of V containing U and W in terms of inclusion.

Definition 1.4 – Linear map

Let V, W be $\mathbb{F}$ -vector spaces, a linear map from V to W is a function $\alpha : V \rightarrow W$ such that for all $u, v \in V$ and $\lambda \in \mathbb{F}$, we have

- (i) $\alpha(u + v) = \alpha(u) + \alpha(v)$;
- (ii) $\alpha(\lambda v) = \lambda \alpha(v)$.

Equivalently, for all $u, v \in V$ and $\lambda \in \mathbb{F}$, $\alpha(u + \lambda v) = \alpha(u) + \lambda \alpha(v)$.

Example 1.3

- (i) Let $V = \mathbb{F}^n, W = \mathbb{F}^m$ and $A \in \mathcal{M}_{m \times n}(\mathbb{F})$. Let $\alpha : V \rightarrow W$ be given by $\alpha(v) = Av$, then α is linear.
- (ii) $\alpha : C^\infty(\mathbb{R}) \rightarrow C^\infty(\mathbb{R})$ with $\alpha(f) = f'$ is linear.
- (iii) $\alpha : C(\mathbb{R}) \rightarrow \mathbb{R}$ with $\alpha(f) = \int_0^1 f(x)dx$ is linear.
- (iv) Let X be any nonempty set and $x_0 \in X$, then the evaluation at x_0 , $\alpha : \mathbb{F}^X \rightarrow \mathbb{F}$ with $\alpha(f) = f(x_0)$ is linear.
- (v) The identity map $\text{id}_V : V \rightarrow V$ with $\text{id}_V(v) = v$ and the zero map $0 : V \rightarrow W$ with $0(v) = 0$ are

linear.

(vi) If $\alpha : V \rightarrow W$ and $\beta : U \rightarrow V$ are both linear, then $\alpha \circ \beta : U \rightarrow W$ is linear.

(vii) $\alpha, \beta : \mathbb{R} \rightarrow \mathbb{R}$ with $\alpha(x) = x^2$ and $\beta(x) = x + 1$ are not linear.

Is $\alpha : \mathbb{C} \rightarrow \mathbb{C}$ with $\alpha(z) = \bar{z}$ a linear map? It depends on the field. If $\mathbb{C}$ is real vector space ($\lambda \in \mathbb{R}$), then yes. If it is a complex vector space ($\lambda \in \mathbb{C}$), then no.

Definition 1.5 – Isomorphism

Let V, W be $\mathbb{F}$ -vector spaces. An isomorphism $\alpha : V \rightarrow W$ is a bijective linear map. We say V and W are isomorphic if there exists an isomorphism $V \rightarrow W$, and we write $V \cong W$.

Example 1.4

Let $V = \mathbb{F}^4$ and $W = \mathcal{M}_{2 \times 2}(\mathbb{F})$. Define $\alpha : V \rightarrow W$ by

$$\begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} \mapsto \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

then α is an isomorphism.

Proposition 1.2

If $\alpha : V \rightarrow W$ is an isomorphism, then so is $\alpha^{-1} : W \rightarrow V$.

Proof. Clearly, α^{-1} is bijective.

For linearity, let $w_1, w_2 \in W$ and $\lambda \in \mathbb{F}$. For $i = 1, 2$, there exists $v_i \in V$ such that $w_i = \alpha(v_i)$. Then,

$$\begin{aligned} \alpha^{-1}(w_1 + \lambda w_2) &= \alpha^{-1}(\alpha(v_1) + \lambda \alpha(v_2)) \\ &= \alpha^{-1}(\alpha(v_1 + \lambda v_2)) \\ &= v_1 + \lambda v_2 \\ &= \alpha^{-1}(w_1) + \lambda \alpha^{-1}(w_2). \end{aligned}$$

□

Definition 1.6 – Kernel and image

Let V, W be $\mathbb{F}$ -vector spaces and $\alpha : V \rightarrow W$ a linear map. The kernel of α is

$$\ker(\alpha) = \{v \in V : \alpha(v) = \mathbf{0}_W\} \subseteq V.$$

The image of α is

$$\operatorname{im}(\alpha) = \{\alpha(v) : v \in V\} \subseteq W.$$

Lemma 1.2

- (i) $\ker(\alpha) \leq V$ and $\text{im}(\alpha) \leq W$.
- (ii) α is surjective $\iff \text{im}(\alpha) = W$.
- (iii) α is injective $\iff \ker(\alpha) = \{\underline{0}_V\}$.

Proof. Check (i) by the subspace test and (ii) is immediate from the definition.

For (iii), suppose α is injective, we know $\alpha(\underline{0}_V) = \alpha(\underline{0}_V + \underline{0}_V) = \alpha(\underline{0}_V) + \alpha(\underline{0}_V)$ so $\alpha(\underline{0}_V) = \underline{0}_W$ and $\underline{0}_V \in \ker(\alpha)$. If $v \in \ker(\alpha)$, then $\alpha(v) = \underline{0}_W = \alpha(\underline{0}_V)$, so $v = \underline{0}_V$ and $\ker(\alpha) = \{\underline{0}_V\}$.

Conversely, suppose $\ker(\alpha) = \{\underline{0}_V\}$. Let $u, v \in V$ and suppose $\alpha(u) = \alpha(v)$. Then, $\alpha(u - v) = \alpha(u) - \alpha(v) = \underline{0}_W$, so $u - v \in \ker(\alpha) = \{\underline{0}_V\}$. Hence, $u = v$ and α is injective. $\square$

Definition 1.7 – Quotient space

Let V be a $\mathbb{F}$ -vector space and $W \leq V$, the quotient space of V by W is

$$V/W = \{v + W : v \in V\},$$

that is the set of left cosets of W in V .

Proposition 1.3

V/W is a $\mathbb{F}$ -vector space under the operations

- (i) $(u + W) + (v + W) = (u + v) + W$;
- (ii) $\lambda(v + W) = \lambda v + W$.

Proof. We only provide a sketch of the proof here.

We first check that the operations are well-defined, that is for $u, \bar{u}, v, \bar{v} \in V$ and $\lambda \in \mathbb{R}$, if $u + W = \bar{u} + W$ and $v + W = \bar{v} + W$, then $(u + v) + W = (\bar{u} + \bar{v}) + W$ and $(\lambda v) + W = (\lambda \bar{v}) + W$. We use the fact that $u + W = v + W$ if and only if $u - v \in W$.

The vector space axioms are inherited from V . $\square$

Definition 1.8 – Quotient map

The function

$$\begin{aligned} \pi_W : V &\rightarrow V/W \\ v &\mapsto v + W \end{aligned}$$

is a surjective linear map with $\ker(\pi_W) = W$, and we call π_W the quotient map.

Proof. Surjectivity is obvious.

For linearity, let $u, v \in V$ and $\lambda \in \mathbb{F}$,

$$\begin{aligned}\pi_W(u + \lambda v) &= (u + \lambda v) + W \\ &= (u + W) + ((\lambda v) + W) \\ &= (u + W) + \lambda(v + W) \\ &= \pi_W(u) + \lambda\pi_W(v).\end{aligned}$$

For the kernel, for $v \in V$,

$$\begin{aligned}v \in \ker(\pi_W) &\iff \pi_W(v) = \underline{0}_{V/W} = \underline{0}_V + W \\ &\iff v + W = \underline{0}_V + W \\ &\iff v = v - \underline{0}_V \in W.\end{aligned}$$

□

Theorem 1.1 – First isomorphism theorem

Let V, W be $\mathbb{F}$ -vector spaces and $\alpha : V \rightarrow W$ be a linear map. Then, there is an isomorphism

$$\begin{aligned}\bar{\alpha} : V/\ker(\alpha) &\rightarrow \text{im}(\alpha) \\ v + \ker(\alpha) &\mapsto \alpha(v).\end{aligned}$$

Proof. Write $K = \ker(\alpha)$. For $u, v \in V$,

$$\begin{aligned}u + K = v + K &\iff u - v \in K \\ &\iff \alpha(u - v) = \underline{0}_W \\ &\iff \alpha(u) = \alpha(v) \\ &\iff \bar{\alpha}(u + K) = \bar{\alpha}(v + K)\end{aligned}$$

so α is well-defined and injective.

For surjectivity, given $w \in \text{im}(\alpha)$, there exists $v \in V$ such that $w = \alpha(v)$, then $w = \bar{\alpha}(v + K)$.

For linearity, given $u, v \in V$ and $\lambda \in \mathbb{F}$,

$$\begin{aligned}\bar{\alpha}((u + K) + \lambda(v + K)) &= \bar{\alpha}((u + \lambda v) + K) \\ &= \alpha(u + \lambda v) \\ &= \alpha(u) + \lambda\alpha(v) \\ &= \bar{\alpha}(u + K) + \lambda\bar{\alpha}(v + K)\end{aligned}$$

□

Definition 1.9 – Span

Let V be a $\mathbb{F}$ -vector space and $S \subseteq V$. The span of S in V is

$$\langle S \rangle = \left\{ \sum'_{s \in S} \lambda_s s : \lambda_s \in \mathbb{F} \right\}$$

where $\sum'$ denotes a finite sum, that is if $|S| = \infty$, then $\lambda_s = 0$ for all but finitely many $s \in S$.

An expression $\sum'_{s \in S} \lambda_s s$ is called a linear combination of S . We say S spans V , or S is a spanning set for V , if $\langle S \rangle = V$. We say V is finite-dimensional if it has a finite spanning set.

- (i) $\langle S \rangle \leq V$. Conversely, if $W \leq V$ and $S \subseteq W$, then $\langle S \rangle \leq W$
- (ii) If $S, T \subseteq V$ and S spans V and $S \subseteq \langle T \rangle$, then T spans V
- (iii) $\langle \emptyset \rangle = \{\underline{0}_V\}$
- (iv) $\langle S \cup T \rangle = \langle S \rangle + \langle T \rangle$

Example 1.5

- (i) Let $V = \mathbb{R}^3$,

$$S = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \right\} \text{ and } T = \left\{ \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix} \right\}.$$

Then,

$$\langle S \rangle = \langle T \rangle = \left\{ \begin{pmatrix} x \\ y \\ 2y \end{pmatrix} : x, y \in \mathbb{R} \right\} \leq \mathbb{R}^3.$$

- (ii) Let $V = \mathbb{R}^{\mathbb{N}}$. Set $T = \{\delta_n : n \in \mathbb{N}\}$ where

$$\delta_n(m) = \delta_{n,m} = \begin{cases} 1 & \text{if } n = m \\ 0 & \text{if } n \neq m. \end{cases}$$

Since the spans only covers a finite sum, $\langle T \rangle \neq \mathbb{R}^{\mathbb{N}}$. In fact,

$$\langle T \rangle = \{f \in \mathbb{R}^{\mathbb{N}} : f(n) = 0 \text{ for all but finitely many } n \in \mathbb{N}\}.$$

Definition 1.10 – Linear independence

Let V be a $\mathbb{F}$ -vector space and $S \subseteq V$. S is linearly independent if, for all linear combinations $\sum'_{s \in S} \lambda_s s$ of S ,

$$\text{if } \sum'_{s \in S} \lambda_s s = \underline{0} \text{ then } \lambda_s = 0 \text{ for all } s \in S.$$

If S is not linearly independent, then it is linearly dependent.

- (i) If $\underline{0} \in S$, then S is linearly dependent.
- (ii) If $S = \{v_1, \dots, v_n\}$ is finite, then S is linearly independent if and only if, for all $\lambda_1, \dots, \lambda_n \in \mathbb{F}$, if $\sum_{i=1}^n \lambda_i v_i = 0$, then $\lambda_1 = \dots = \lambda_n = 0$.
- (iii) If $|S| = \infty$, then S is linearly independent if and only if every finite subset of S is linearly independent.
- (iv) Every subset of a linearly independent set is linearly independent.

Example 1.6

- (i)

$$\begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} - 5 \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} + 2 \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix} = \underline{0}$$

so T is linearly independent. But, for all $\lambda, \mu \in \mathbb{R}$, if

$$\begin{pmatrix} \lambda + \mu \\ \mu \\ 2\mu \end{pmatrix} = \lambda \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \underline{0},$$

then $\lambda = \mu = 0$, so S is linearly dependent.

(ii) If $f = \sum_{n \in \mathbb{N}}' \lambda_n \delta_n = \underline{0}$, then for all $m \in \mathbb{N}$,

$$0 = f(m) = \sum_{n \in \mathbb{N}}' \lambda_n \delta_n(m) = \lambda_m,$$

so T is linearly independent.

Definition 1.11 – Basis

Let $S \subseteq V$. S is a basis for V if S is linearly independent and spans V .

Example 1.7

- (i) Let $e_i \in \mathbb{F}^n$ be given by $e_i = (0 \dots 0 \ 1 \ 0 \dots 0)$ where 1 is in the i -th entry. Then, $\{e_i : 1 \leq i \leq n\}$ is the standard basis for $\mathbb{F}^n$.
- (ii) $\{(1 \ 0 \ 0), (1 \ 1 \ 0), (1 \ 1 \ 1)\}$ is a basis for $\mathbb{F}^3$.
- (iii) Recall that $P(\mathbb{R})$ is the set of real polynomial functions. Let $p_n \in P(\mathbb{R})$ be given by $p_n(x) = x^n$. Then, $\{p_n : n \in \mathbb{Z}_{\geq 0}\}$ is a basis for $P(\mathbb{R})$.

Proposition 1.4

Let $S \subseteq V$ be a finite spanning set. There exists $S' \subseteq S$ such that S' is a basis for V .

Proof. If S is linearly independent, then we are done.

If not, write $S = \{v_1, \dots, v_n\}$, then there exists $\lambda_1, \dots, \lambda_n \in \mathbb{F}$ such that $\sum_{i=1}^n \lambda_i v_i = \underline{0}$ with (without loss of generality) $\lambda_n \neq 0$. Then,

$$v_n = -\frac{1}{\lambda_n} \sum_{i=1}^{n-1} \lambda_i v_i$$

so $v_n \in \langle S \setminus \{v_n\} \rangle$. Hence, $S \setminus \{v_n\}$ spans S and so spans V . Replace S by $S \setminus \{v_n\}$ and repeat. This terminates since S is finite and we are left with a linearly independent set. $\square$

Definition 1.12 – Steinitz exchange lemma

Let $S, T \subseteq V$ be finite with S linearly independent and T a spanning set for V . Then,

- (i) $|S| \leq |T|$;
- (ii) there exists $T' \subseteq T$ such that $|T'| = |T| - |S|$ and $S \cup T'$ spans V .

Proof. Let $S = \{u_1, \dots, u_m\}$ and $T = \{v_1, \dots, v_n\}$. If $S = \emptyset$, then we are done. If not, there exists $\lambda_i \in \mathbb{F}$ such

that $u_1 = \sum_{i=1}^n \lambda_i v_i$ with, by relabelling, $\lambda_1 \neq 0$. Then,

$$v_1 = \frac{1}{\lambda_1} \left(u_1 - \sum_{i=2}^n \lambda_i v_i \right)$$

so $\{u_1, v_2, \dots, v_n\}$ spans V . After k steps with $k < m$, $\{u_1, \dots, u_k, v_{k+1}, \dots, v_n\}$ spans. There exists $\lambda_i, \mu_j \in \mathbb{F}$ such that

$$u_{k+1} = \sum_{j=1}^k \mu_j u_j + \sum_{i=k+1}^n \lambda_i v_i.$$

If all $\lambda_i = 0$, then

$$\left(\sum_{j=1}^k \mu_j u_j \right) - u_{k+1} = \underline{0}$$

which is a contradiction since S is linearly independent. So, by relabelling, $\lambda_{k+1} \neq 0$ and

$$v_{k+1} = \frac{1}{\lambda_{k+1}} \left(u_{k+1} - \sum_{j=1}^k \mu_j u_j - \sum_{i=k+2}^n \lambda_i v_i \right)$$

so $\{u_1, \dots, u_{k+1}, v_{k+2}, \dots, v_n\}$ spans V . We conclude that $m \leq n$ and $\{u_1, \dots, u_m, v_{m+1}, \dots, v_n\}$ spans V , hence $T' = \{v_{m+1}, \dots, v_n\}$. $\square$

Corollary 1.1

Let V be a finite-dimensional $\mathbb{F}$ -vector space. Then,

- (i) every basis for V is finite
- (ii) all finite bases for V have the same size

Proof.

- (i) Let the finite basis of V be B . Suppose B' is an infinite basis, let $B'' \subseteq B'$ with $|B''| = |B| + 1$, then B'' is linearly independent. Applying Steinitz (i) with $S = B''$ and $T = B$, we get a contradiction.
- (ii) Let $B_1, B_2 \subseteq V$ be finite bases. Apply Steinitz (i) twice, taking $S = B_1$ and $T = B_2$, we get $|B_1| \leq |B_2|$, swapping them, we get $|B_2| \leq |B_1|$. So, $|B_1| = |B_2|$. $\square$

Definition 1.13 – Dimension

The dimension $\dim(V)$ of a finite-dimensional vector space V is the size of any basis.

This is well-defined by the above proof.

Example 1.8

- (i) $\dim(\mathbb{F}^n) = n$.
- (ii) $\mathbb{C}$ is one-dimensional as a complex vector space but two-dimensional as a real vector space with $\{1, i\}$ being a basis over $\mathbb{R}$.

Corollary 1.2

Let $S, T \subseteq V$ be finite with S linearly independent and T a spanning set of V . Then,

$$|S| \leq \dim(V) \leq |T|.$$

$|S| = \dim(V)$ if and only if it is a basis, and similarly for T .

Proof. The inequalities are immediate from Steinitz (i).

If S is a basis, then $|S| = \dim(V)$ by Corollary 1.1. Conversely, suppose $|S| = \dim(V)$ and let B be a basis for V so $|B| = |S|$. Then, B spans V and we can apply Steinitz (ii). So, there exists $B' \subseteq B$ such that $|B'| = |B| - |S| = 0$ and $S = S \cup B'$ spans V . Hence, S is a basis. The proof for T is similar. $\square$

A warning, we shall not prove that every vector space has a basis. This is connected to the axiom of choice. However, some non-finite-dimensional vector spaces do have a basis.

Proposition 1.5

If V is finite-dimensional and $U \leq V$, then U is also finite-dimensional with $\dim(U) \leq \dim(V)$, with equality if and only if $U = V$.

Proof. If $U = \{0_V\}$, then we are done. If not, pick some $u_1 \in U$ with $u_1 \neq 0_V$. Then, $\{u_1\} \subseteq U$ is linearly independent. After k steps, we have $\{u_1, \dots, u_k\}$ being linearly independent and $k \leq \dim(V)$. If $\{u_1, \dots, u_k\}$ spans U , then we are done.

If not, let $u_{k+1} \in U \setminus \langle u_1, \dots, u_k \rangle$. If $\{u_1, \dots, u_{k+1}\}$ is not linearly independent, then $\sum_i \lambda_i u_i = 0$ with $\lambda_{k+1} \neq 0$. Thus,

$$u_{k+1} = -\frac{1}{\lambda_{k+1}} \sum_{i=1}^k \lambda_i u_i \in \langle u_1, \dots, u_k \rangle,$$

hence contradiction. This terminates after $\leq \dim(V)$ steps by Corollary 1.2.

If $\dim(U) = \dim(V)$, apply Corollary 1.2 with S being any basis for U . $\square$

Proposition 1.6

Let V be finite-dimensional and $U \leq V$. For any basis of B_U of U , there exists a basis B_V of V such that $B_U \subseteq B_V$.

Proof. Apply Steinitz (ii) with $S = B_U$ and T being any basis for V . We obtain $T' \subseteq T$ such that $|T'| = |T| - |S| = \dim(V) - \dim(U)$ and $B_V = B_U \cup T'$ spans V . But $|B_V| \leq |B_U| + |T'| = \dim(V)$, so B_V is a basis for V . $\square$

Theorem 1.2 – Rank-nullity theorem

Let V be a finite-dimensional $\mathbb{F}$ -vector space, W be a $\mathbb{F}$ -vector space and $\alpha : V \rightarrow W$ be a linear map. Then, $\text{im}(\alpha)$ is finite-dimensional and

$$\dim(V) = \dim(\ker(\alpha)) + \dim(\text{im}(\alpha)).$$

The nullity of α is $n(\alpha) = \dim(\ker(\alpha))$ and the rank of α is $\text{rk}(\alpha) = \dim(\text{im}(\alpha))$.

Proof. By the first isomorphism theorem, $V/\ker(\alpha) \cong \text{im}(\alpha)$, so it is sufficient to prove

Lemma 1.3

Let $U \leq V$, then $\dim(V/U) = \dim(V) - \dim(U)$.

Proof. Let $B_U = \{u_1, \dots, u_m\}$ be a basis of U . Extend it to a basis $B_V = \{u_1, \dots, u_m, v_{m+1}, \dots, v_n\}$ of V . Set $B_{V/U} = \{v_i + U : m+1 \leq i \leq n\}$, we claim that this is a basis of size $n - m$ for V/U .

We first check that it spans V/U . For $v \in V$, write $v = \sum_i \lambda_i v_i + \sum_j \mu_j u_j$, then $v + U = \sum_i \lambda_i (v_i + U) = \langle B_{V/U} \rangle$.

For linear independence, suppose $\sum_i \lambda_i (v_i + U) = \underline{0} + U$. But $\sum_i \lambda_i (v_i + U) = (\sum_i \lambda_i v_i) + U$ so $\sum_i \lambda_i v_i \in U$, that is $\sum_i \lambda_i v_i = \sum_j \mu_j u_j$ for some $\mu_j \in \mathbb{F}$. Since B_V is linearly independent, all $\lambda_i, \mu_j = 0$. Similarly, if $v_i + U = v_j + U$ with $i \neq j$, then we can write $v_i - v_j = \sum_j \mu_j u_j$ which is a contradiction. $\square$

This directly implies the rank-nullity theorem. $\square$

We can also prove the rank-nullity theorem avoiding quotient spaces. Start with a basis B_K for $\ker(\alpha)$, extend to a basis B_V for V and check that $\alpha(B_V \setminus B_K)$ is a basis for $\text{im}(\alpha)$.

Corollary 1.3 – Linear pigeonhole

If $\dim(V) = \dim(W) = n$ and $\alpha : V \rightarrow W$ is a linear map, then the following are equivalent

- (i) α is injective;
- (ii) α is surjective;
- (iii) α is an isomorphism.

Proof. (i) is true iff $n(\alpha) = 0$ and (ii) is true iff $\text{rk}(\alpha) = n$. By the rank-nullity theorem, either of these implies the other. $\square$

Proposition 1.7

Let V be a vector space, not necessarily finite-dimensional. Suppose we have a basis B for V . For any vector space W and any function $f : B \rightarrow W$, there is a unique linear map $F : V \rightarrow W$ such that $F|_B = f$.

Proof. First we prove existence, for $v \in V$, write $v = \sum'_b \lambda_b b$ with $\lambda_b \in \mathbb{F}$. Define $F(v) = \sum'_b \lambda_b f(b)$. This is well-defined as B is a basis, so the λ_b are uniquely determined by v .

This is linear, for $u, v \in V$ and $\lambda \in \mathbb{F}$, write $u = \sum'_b \mu_b b$ and $v = \sum'_b \lambda_b b$. Then,

$$\begin{aligned} F(u + \lambda v) &= F\left(\sum'_b (\mu_b + \lambda \lambda_b) b\right) \\ &= \sum'_b (\mu_b + \lambda \lambda_b) f(b) \\ &= \sum'_b \mu_b f(b) + \lambda \sum'_b \lambda_b f(b) \\ &= F(u) + \lambda F(v). \end{aligned}$$

For uniqueness, if $\bar{F} : V \rightarrow W$ is another linear map extending f , then

$$\bar{F} \left(\sum'_b \lambda_b b \right) = \sum'_b \lambda_b \bar{F}(b) = \sum'_b \lambda_b f(b) = F \left(\sum'_b \lambda_b b \right).$$

so $\bar{F} = F$. □

Corollary 1.4

Let $\dim(V) = n$ and $B = \{v_1, \dots, v_n\}$ be a basis of V , then there is an isomorphism

$$F_B : V \rightarrow \mathbb{F}^n$$

$$\sum_{i=1}^n \lambda_i v_i \mapsto \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix}.$$

$F_B(v) = [v]_B$ is the co-ordinate vector of V with respect to B .

Proof. Let $E = \{e_1, \dots, e_n\}$ be the standard basis for $\mathbb{F}^n$. Define $f : B \rightarrow \mathbb{F}^n$ by $f(v_i) = e_i$. Let F_B be the unique linear extension of f to V . f defines a bijection from B to E . Let $\bar{F}_B$ be the linear extension of $f^{-1} : E \rightarrow B$ to $\mathbb{F}^n$. Then, $\bar{F}_B \circ F_B$ is a linear extension of id_V . But id_V is also a linear extension of id_B , so $\bar{F}_B \circ F_B = \text{id}_V$. Similarly, $F_B \circ \bar{F}_B = \text{id}_{\mathbb{F}^n}$, so $\bar{F}_B = F_B^{-1}$ and F_B is an isomorphism. □

Corollary 1.5

Let V, W be finite-dimensional $\mathbb{F}^n$ -vector spaces. Then, $V \cong W$ if and only if $\dim(V) = \dim(W)$.

Proof. If $\dim(V) = \dim(W) = n$, then by Corollary 1.4, $V \cong \mathbb{F}^n \cong W$. Conversely, the image of a basis under an isomorphism is also a basis. □

Example 1.9

If $V \cong \mathbb{F}^n$ and $U \leq V$ with $U \cong \mathbb{F}^m$, then $\dim(V/U) = n - m$ so $V/U \cong \mathbb{F}^{n-m}$.

Definition 1.14 – (External) direct sum

Let V and W be $\mathbb{F}$ -vector spaces. The (external) direct sum of V and W , denoted $V \oplus W$, is the $\mathbb{F}$ -vector space with underlying set $V \times W$ and operations $(v_1, w_1) + (v_2, w_2) = (v_1 + w_1, v_2 + w_2)$ and $\lambda(v, w) = (\lambda v, \lambda w)$.

It is easy to check that it satisfies the vector space axioms. Similarly, we can define $V_1 \oplus \dots \oplus V_n$.

Lemma 1.4

Let V, W be finite-dimensional, then $\dim(V \oplus W) = \dim(V) + \dim(W)$.

Proof. Let B, C be bases for V, W respectively. Set $D = (B \times \{0_W\}) \cup (\{0_V\} \times C)$. Then, it is straightforward to check that D is a basis with $|D| = |B| + |C|$. Alternatively, suppose $V \cong \mathbb{F}^n$ and $W \cong \mathbb{F}^m$. We can then construct an isomorphism $V \oplus W \cong \mathbb{F}^{n+m}$. □

Proposition 1.8

Let V be a $\mathbb{F}$ -vector space and $U, W \leq V$. Then, there is a surjective linear map

$$\begin{aligned}\phi : U \oplus W &\rightarrow U + W \\ (u, w) &\mapsto u + w\end{aligned}$$

with $\ker(\phi) \cong U \cap W$.

Proof. Surjectivity and linearity is clear. For the kernel, note that for $(u, w) \in U \oplus W$, $(u, w) \in \ker(\phi)$ if and only if $w = -u$. Hence, $\ker(\phi) = \{(x, -x) : x \in U \cap W\}$. The map

$$\begin{aligned}\psi : U \cap W &\rightarrow \ker(\phi) \\ x &\mapsto (x, -x)\end{aligned}$$

is an isomorphism, which gives the result. $\square$

Corollary 1.6 – Sum-intersection formula

If V is finite-dimensional and $U, W \leq V$, then

$$\dim(U + W) = \dim(U) + \dim(W) - \dim(U \cap W).$$

Proof. Apply the rank-nullity theorem to ϕ from Proposition 1.8.

$$\begin{aligned}\dim(U) + \dim(W) &= \dim(U \oplus W) \\ &= \dim(\ker(\phi)) + \dim(\operatorname{im}(\phi)) \\ &= \dim(U \cap W) + \dim(U + W).\end{aligned}$$

$\square$

Given a basis B for $U \cap W$. We can extend B to a basis B_U for U and to a basis B_W for W . Then, $B_U \cup B_W$ spans $U + W$ and $|B_U \cup B_W| \leq |B_U| + |B_W| - |B|$. By the sum-intersection formula, $|B_U \cup B_W| = \dim(U + W)$. Hence, $B_U \cup B_W$ is linearly independent, and is a basis. We could have also checked this directly, giving an alternative proof of the sum-intersection formula.

Let V be a vector space. Suppose $U, W \leq V$ satisfy $U + W = V$ and $U \cap W = \{0_V\}$. Then, by Proposition 1.8, $\phi : U \oplus W \rightarrow V$ is an isomorphism. If U, W satisfy the two conditions, we say V is the (internal) direct sum of U and W and we write $V = U \oplus W$. Alternatively, we can say every element $v \in V$ can be written uniquely as $v = u + w$ for $u \in U$ and $w \in W$. So,

Definition 1.15 – (Internal) direct sum

Let V be a vector space and suppose $U, W \leq V$ satisfies

- (i) $U + W = V$
- (ii) $U \cap W = \{0_V\}$,

then $V = U \oplus W$. Equivalently, every element $v \in V$ can be written uniquely as $v = u + w$ for $u \in U$ and $w \in W$.

Definition 1.16 – Direct complement

For $U \leq V$, a direct complement to U in V is a subspace $W \leq V$ satisfying $V = U \oplus W$.

Proposition 1.9

If V is finite-dimensional, then every subspace has a direct complement.

Proof. Let $U \leq V$ and B_U be a basis for U . Extend to a basis B_V for V . Set $W = \langle B_V \setminus B_U \rangle$. Then,

$$\begin{aligned} V &= \langle B_V \rangle \\ &= \langle B_U \cup (B_V \setminus B_U) \rangle \\ &= \langle B_U \rangle + \langle B_V \setminus B_U \rangle = U + W. \end{aligned}$$

Moreover, as B_V is linearly independent, $U \cap W = \langle B_U \rangle \cap \langle B_V \setminus B_U \rangle = \{0\}$. Alternatively, by the sum-intersection formula, $\dim(U \cap W) = 0$. $\square$

More generally, for $U_1, \dots, U_n \leq V$, we say V is the direct sum of the U_i , and write $V = U_1 \oplus \dots \oplus U_n$, if the map

$$\begin{aligned} \phi : U_1 \oplus \dots \oplus U_n &\rightarrow V \\ (u_1, \dots, u_n) &\mapsto u_1 + \dots + u_n \end{aligned}$$

is an isomorphism. Equivalently, $V = U_1 \oplus \dots \oplus U_n$ if and only if every $v \in V$ can be uniquely written as $v = u_1 + \dots + u_n$ with $u_i \in U_i$. Note that we need to be careful when generalising the definitions, $V = U_1 \oplus \dots \oplus U_n$ is not equivalent to $V = U_1 + \dots + U_n$ and $U_i \cap U_j = \{0\}$ for all $i \neq j$.

2 Matrices and linear maps

Let V, W be $\mathbb{F}$ -vector spaces and

$$\mathcal{L}(V, W) = \{\alpha : V \rightarrow W \text{ such that } \alpha \text{ is linear}\}.$$

For $\alpha, \beta \in \mathcal{L}(V, W)$ and $\lambda \in \mathbb{F}$, define $(\alpha + \beta), (\lambda\alpha) \in \mathcal{L}(V, W)$ by $(\alpha + \beta)(v) = \alpha(v) + \beta(v)$ and $(\lambda\alpha)(v) = \lambda\alpha(v)$. We can check that both are linear. Also, $\mathcal{L}(V, W)$ is a $\mathbb{F}$ -vector space under these operations.

Theorem 2.1

Let V, W be finite-dimensional $\mathbb{F}$ -vector spaces with $\dim(V) = n$ and $\dim(W) = m$. Then,

$$\mathcal{L}(V, W) \cong \mathcal{M}_{m \times n}(\mathbb{F}).$$

In particular, $\dim(\mathcal{L}(V, W)) = mn$.

Recall that $\mathcal{M}_{m \times n}(\mathbb{F})$ is the space of matrices over $\mathbb{F}$, with m rows and n columns. For $A \in \mathcal{M}_{m \times n}(\mathbb{F})$, we can write $A = (a_{ij})$ with $a_{ij} \in \mathbb{F}$, $1 \leq i \leq m$ and $1 \leq j \leq n$.

Let $B = \{v_1, \dots, v_n\}$ and $C = \{w_1, \dots, w_m\}$ be ordered basis for V and W . Let $\alpha \in \mathcal{L}(V, W)$ and write

$$\begin{aligned} \alpha(v_1) &= a_{11}w_1 + a_{21}w_2 + \dots + a_{m1}w_m \\ \alpha(v_2) &= a_{12}w_1 + a_{22}w_2 + \dots + a_{m2}w_m \\ &\vdots \\ \alpha(v_n) &= a_{1n}w_1 + a_{2n}w_2 + \dots + a_{mn}w_m. \end{aligned}$$

The matrix of α with respect to the ordered basis B and C is $[\alpha]_C^B = (a_{ij}) \in \mathcal{M}_{m \times n}(\mathbb{F})$. Note that the rows of the set of equations above become the columns of $[\alpha]_C^B$. Recall from before that there is a linear isomorphism $\varepsilon_B : V \rightarrow \mathbb{F}^n$ that maps vectors v to the co-ordinate vector with respect to a basis $[v]_B$. With basic knowledge equipped, we have the following

Theorem 2.2

- (i) For all $v \in V$, $[\alpha]_C^B[v]_B = [\alpha(v)]_C$.
- (ii) $[\alpha]_C^B$ is the only matrix $A \in \mathcal{M}_{m \times n}(\mathbb{F})$ satisfying $A[v]_B = [\alpha(v)]_C$ for all $v \in V$.
- (iii) There is an isomorphism of $\mathbb{F}$ -vector spaces

$$\begin{aligned} \varepsilon_C^B : \mathcal{L}(V, W) &\rightarrow \mathcal{M}_{m \times n}(\mathbb{F}) \\ \alpha &\mapsto [\alpha]_C^B. \end{aligned}$$

Proof.

- (i) Let $v \in V$ and write $v = \sum_{j=1}^n \lambda_j v_j$. Then,

$$\alpha(v) = \sum_{j=1}^n \lambda_j \alpha(v_j) = \sum_{j=1}^n \lambda_j \sum_{i=1}^m a_{ij} w_i = \sum_{i=1}^m \left(\sum_{j=1}^n \lambda_j a_{ij} \right) w_i$$

so

$$[\alpha(v)]_C = \left(\sum_{j=1}^n a_{ij} \lambda_j \right)_i = (a_{ij}) \cdot (\lambda_i) = [\alpha]_C^B [v]_B.$$

- (ii) We know that $[v_j]_B = e_j$ for $1 \leq j \leq n$. Hence, for $A \in \mathcal{M}_{m \times n}(\mathbb{F})$, $A[v_j]_B = j$ -th column of A . But if

$$A[v_j]_B = [\alpha(v_j)]_C = [\alpha]_C^B [v_j]_B = [\alpha]_C^B e_j,$$

then $A[v_j]_B$ is also the j -th column of $[\alpha]_C^B$.

- (iii) We first check linearity. Let $\alpha, \beta \in \mathcal{L}(V, W)$ and $\lambda \in \mathbb{F}$, then

$$[\alpha + \lambda\beta]_C^B [v]_B = [(\alpha + \lambda\beta)(v)]_C = [\alpha(v) + \lambda\beta(v)]_C = [\alpha(v)]_C + \lambda[\beta(v)]_C = ([\alpha]_C^B + \lambda[\beta]_C^B)[v]_B$$

for all $v \in V$. Hence, by (ii), $[\alpha + \lambda\beta]_C^B = [\alpha]_C^B + \lambda[\beta]_C^B$.

For injectivity, let $\alpha \in \ker(\varepsilon_C^B)$ so that $[\alpha]_C^B = 0 \in \mathcal{M}_{m \times n}(\mathbb{F})$. Then, by (i), $[\alpha(v)]_C = \mathbf{0}$ for all $v \in V$. But $\varepsilon_C : w \mapsto [w]_C$ is an isomorphism so $\alpha(v) = \mathbf{0}$ for all $v \in V$, that is $\alpha = 0$. Lastly, for surjectivity, let $A \in \mathcal{M}_{m \times n}(\mathbb{F})$. Define $f : B \rightarrow W$ by $f(v_j) = \sum_{i=1}^m a_{ij} w_i$ and extend f to a linear map $F : V \rightarrow W$. Then, $[F]_C^B = A$.

□

Proposition 2.1

Let V, W, X be finite-dimensional $\mathbb{F}$ -vector spaces with basis B, C, D respectively. Let $\alpha \in \mathcal{L}(V, W)$ and $\beta \in \mathcal{L}(W, X)$, then

$$[\beta \circ \alpha]_D^B = [\beta]_D^C [\alpha]_C^B$$

Proof. By Theorem 2.2, $[\beta \circ \alpha]_D^B$ is the unique matrix A satisfying $A[v]_B = [\beta(\alpha(v))]_D$ for all $v \in V$. But

$$[\beta]_D^C [\alpha]_C^B [v]_B = [\beta]_D^C [\alpha(v)]_C = [\beta(\alpha(v))]_D,$$

so by (ii) of Theorem 2.2, they are equal.

□

For any basis B of V , $[\text{id}_V]_B^B = I_{\dim(V)}$.

Definition 2.1 – Change-of-basis matrix

Let B, B' be basis for V with $\dim(V) = n$. The change-of-basis matrix from B to B' is given by

$$P = [\text{id}_V]_{B'}^B \in \mathcal{M}_{n \times n}(\mathbb{F}).$$

Equivalently, letting $B = \{v_1, \dots, v_n\}$ and $B' = \{v'_1, \dots, v'_n\}$, we get

$$P = (p_{ij}) \text{ where } v_j = \sum_{i=1}^n p_{ij} v'_i.$$

So, the j -th column of P is $[v_j]_{B'}$.

Recall that $A \in \mathcal{M}_{n \times n}(\mathbb{F})$ is invertible if there exists $A^{-1} \in \mathcal{M}_{n \times n}(\mathbb{F})$ such that $AA^{-1} = A^{-1}A = I_n$. A^{-1} is called an inverse for A . $\text{GL}_n(\mathbb{F})$ denotes the set (group) of $n \times n$ invertible matrices over $\mathbb{F}$.

Proposition 2.2

- (i) $[\text{id}_V]_{B'}^B \in \text{GL}_n(\mathbb{F})$ with inverse $[\text{id}_V]_B^{B'}$.
- (ii) Let $\alpha \in \mathcal{L}(V, W)$ with B, B' being basis for V and C, C' being basis for W . Then,

$$[\alpha]_{C'}^{B'} = [\text{id}_W]_{C'}^C [\alpha]_C^B [\text{id}_V]_B^{B'}.$$

Proof. For (i), we know that $I_n = [\text{id}_V]_B^B = [\text{id}_V]_B^{B'} [\text{id}_V]_{B'}^B$ and similarly by swapping B and B' . (ii) is immediate from the proposition. $\square$

Definition 2.2 – Equivalent Matrices

$A, A' \in \mathcal{M}_{m \times n}(\mathbb{F})$ are equivalent if there exists $P \in \text{GL}_m(\mathbb{F}), Q \in \text{GL}_n(\mathbb{F})$ such that

$$A' = PAQ.$$

- (i) $[\alpha]_C^B$ and $[\alpha]_{C'}^{B'}$ are equivalent.
- (ii) A is equivalent to A .
- (iii) If $A' = PAQ$, then $A = P^{-1}A'Q^{-1}$.
- (iv) If $A' = PAQ$ and $A'' = RA'S$, then $A'' = (RP)A(QS)$.
- (v) The equivalence of matrices is an equivalence relation on $\mathcal{M}_{m \times n}(\mathbb{F})$.

Here, we use square brackets to denote block matrices and use $*$ to denote potentially non-zero matrix entries.

Theorem 2.3

Let V, W be finite-dimensional vector spaces with $\dim(V) = n$ and $\dim(W) = m$. Let $\alpha \in \mathcal{L}(V, W)$ and $r = \text{rk}(\alpha)$. Then,

- (i) there exists bases B, C for V, W respectively such that $[\alpha]_C^B = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \in \mathcal{M}_{m \times n}(\mathbb{F})$.
- (ii) if $[\alpha]_{C'}^{B'} = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$ for some bases B', C' for V, W respectively and some r' , then $r = r'$.

Proof.

- (i) By the rank-nullity theorem, $n(\alpha) = n - r$. Let $\{v_{r+1}, \dots, v_n\}$ be a basis for $\ker(\alpha)$ and extend to a basis $B = \{v_1, \dots, v_r, v_{r+1}, \dots, v_n\}$. Then, $\{\alpha(v_1), \dots, \alpha(v_n)\}$ spans $\text{im}(\alpha)$ and has size at most $r = \dim(\text{im}(\alpha))$. Hence, $\{\alpha(v_1), \dots, \alpha(v_r)\}$ is linear independent, so we can extend it to a basis $C = \{w_1 = \alpha(v_1), \dots, w_r = \alpha(v_r), w_{r+1}, \dots, w_m\}$. Then,

$$\alpha(v_j) = \begin{cases} w_j & \text{if } 1 \leq j \leq r \\ \underline{0} & \text{otherwise} \end{cases}$$

and we get the result.

- (ii) If $[\alpha]_{C'}^{B'} = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$, then

$$\alpha(v'_j) = \begin{cases} w'_j & \text{if } 1 \leq j \leq r' \\ \underline{0} & \text{otherwise} \end{cases}.$$

Hence, $\{w'_1, \dots, w'_{r'}\}$ span $\text{im}(\alpha)$ and are linearly independent (as a subset of C'). Thus, $\text{rk}(\alpha) = r'$.

□

Definition 2.3 – (Column/Row)-(space/rank)

For $A \in \mathcal{M}_{m \times n}(\mathbb{F})$, the column-space $\text{Col}(A)$ is the subspace of $\mathbb{F}^m$ spanned by the columns of A . The column-rank of A is $\text{c-rk}(A) = \dim(\text{Col}(A))$. Similarly, we have the row-space $\text{Row}(A)$ and the row-rank $\text{r-rk}(A) = \dim(\text{Row}(A))$.

Note that $\text{Row}(A) \leq \mathbb{F}^n$ (column-vectors of length n). Also, $\text{Row}(A) = \text{Col}(A^\top)$, hence $\text{r-rk}(A) = \text{c-rk}(A^\top)$. Given $A \in \mathcal{M}_{m \times n}(\mathbb{F})$, we can define the linear map

$$\begin{aligned} \alpha : \mathbb{F}^n &\rightarrow \mathbb{F}^m \\ \alpha(v) &\mapsto Av \end{aligned}$$

Then, $\text{im}(\alpha) = \text{Col}(A)$ so $\text{rk}(\alpha) = \text{c-rk}(A)$. Moreover, $A = [\alpha]_{E_m}^{E_n}$ where E_k is the standard basis for $\mathbb{F}^k$. We may sometimes write $\text{im}(A), \ker(A), \dots$ to mean $\text{im}(\alpha), \ker(\alpha), \dots$.

Theorem 2.4

Let $A, A' \in \mathcal{M}_{m \times n}(\mathbb{F})$, then

- (i) A is equivalent to $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$ where $r = \text{c-rk}(A)$.

- (ii) A and A' are equivalent if and only if $\text{c-rk}(A) = \text{c-rk}(A')$.

Proof. This proof requires the following lemma.

Lemma 2.1

For $A \in \mathcal{M}_{m \times n}(\mathbb{F})$ and $B \in \mathcal{M}_{n \times p}(\mathbb{F})$, $\text{rk}(AB) \leq \min(\text{rk}(A), \text{rk}(B))$.

Proof. $\text{im}(AB) \leq \text{im}(A)$ so $\text{rk}(AB) \leq \text{rk}(A)$. Also, if $Bv = \underline{0}$ for some $v \in \mathbb{F}^p$, then $ABv = \underline{0}$ so $n(B) \leq n(AB)$. By the rank-nullity theorem, $p - \text{rk}(B) \leq p - \text{rk}(AB)$ so $\text{rk}(AB) \leq \text{rk}(B)$. □

Now, we prove the theorem.

- (i) Let α be the linear map corresponding to A , so $A = [\alpha]_{E_n}^{E_m}$. By Theorem 2.3, there exist bases B, C for $\mathbb{F}^n, \mathbb{F}^m$ respectively such that

$$\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} = [\alpha]_C^B = [\text{id}_{\mathbb{F}^m}]_C^{E_m} [\alpha]_{E_m}^{E_n} [\text{id}_{\mathbb{F}^n}]_{E_n}^B = PAQ$$

which is equivalent to A .

- (ii) If A' has column-rank r also, then A' is equivalent to $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$ by (i). Hence, A' is equivalent to A . Conversely, if $A' = PAQ$, then by Lemma 2.1, $\text{rk}(A') \leq \text{rk}(AQ) \leq \text{rk}(A)$. And by symmetry, $\text{rk}(A) \leq \text{rk}(A')$, hence proved. $\square$

Theorem 2.5

For any $A \in \mathcal{M}_{m \times n}(\mathbb{F})$, then $\text{r-rk}(A) = \text{c-rk}(A)$.

Proof. Note that if P is invertible, then $P^\top$ is also invertible with inverse $(P^{-1})^\top$. Let $\text{c-rk}(A) = r$, then there exists $P \in \text{GL}_m(\mathbb{F})$ and $Q \in \text{GL}_n(\mathbb{F})$ such that

$$PAQ = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \in \mathcal{M}_{m \times n}(\mathbb{F}).$$

Then $A^\top$ is equivalent to

$$Q^\top A^\top P^\top = (PAQ)^\top = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \in \mathcal{M}_{n \times m}(\mathbb{F}).$$

By Theorem 2.4, $\text{c-rk}(A^\top) = r = \text{r-rk}(A)$. $\square$

Let V be finite-dimensional vector spaces and B, B' be bases for V and $\alpha \in \mathcal{L}(V, W)$. Then, $[\alpha]_{B'}^{B'} = [\text{id}_V]_{B'}^B [\alpha]_B^B [\text{id}_V]_B^{B'} = P^{-1} [\alpha]_B^B P$. With this, we can define

Definition 2.4 – Similar matrices

$A, A' \in \mathcal{M}_{m \times n}(\mathbb{F})$ are similar if there exists $P \in \text{GL}_n(\mathbb{F})$ such that

$$A' = P^{-1}AP$$

- (i) Similarity is an equivalence relation on $\mathcal{M}_{m \times n}(\mathbb{F})$.
- (ii) Similar matrices are equivalent but equivalent matrices need not be similar. For example, I_n is equivalent to every matrix in $\text{GL}_n(\mathbb{F})$, but is only similar to itself.

Definition 2.5 – Elementary row operations

For $A \in \mathcal{M}_{m \times n}(\mathbb{R})$, let $\mathbf{r}_1, \dots, \mathbf{r}_m$ be the rows of A . There are three types of elementary row operations on A :

- (i) swap $\mathbf{r}_i$ and $\mathbf{r}_j$ with $i \neq j$;
- (ii) replace $\mathbf{r}_i$ by $\lambda \mathbf{r}_i$ with $0 \neq \lambda \in \mathbb{F}$;
- (iii) replace $\mathbf{r}_i$ by $\mathbf{r}_i + \lambda \mathbf{r}_j$ with $\lambda \in \mathbb{F}$ and $i \neq j$.

Similarly, there are also three types of elementary column operations. We see that all of these operations are

reversible. Corresponding to these operations, there are elementary matrices.

$$\begin{aligned}
 \text{(i) } T_{i,j} &= \begin{pmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 1 & & & & \\ & & & 0 & & & 1 \\ & & & & 1 & & \\ & & & & & \ddots & \\ & & & & & & 1 \\ & & & & & & & 0 & & & 1 \\ & & & & & & & & 1 & & \\ & & & & & & & & & \ddots & \\ & & & & & & & & & & 1 \end{pmatrix} \text{ which is } I_m \text{ with the } i\text{-th and } j\text{-th row swapped.} \\
 \text{(ii) } M_{i,\lambda} &= \begin{pmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 1 & & & & \\ & & & \lambda & & & \\ & & & & 1 & & \\ & & & & & \ddots & \\ & & & & & & 1 \end{pmatrix} \text{ where the } i\text{-th one in } I_m \text{ has been replaced by a } \lambda. \\
 \text{(iii) } C_{i,j,\lambda} &= \begin{pmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & & \lambda & & & \\ & & & & \ddots & & \\ & & & & & \ddots & \\ & & & & & & 1 \end{pmatrix} \text{ which is } I_m \text{ with the } (i,j) \text{ entry being } \lambda.
 \end{aligned}$$

They are obtained from I_m by applying an elementary row operation and are invertible with $T_{i,j}^{-1} = T_{i,j}$, $M_{i,\lambda}^{-1} = M_{i,\lambda^{-1}}$ and $C_{i,j,\lambda}^{-1} = C_{i,j,-\lambda}$.

Lemma 2.2

If E is an elementary matrix of type (i), (ii) or (iii), then EA is obtained from A by applying the elementary row operation of the corresponding type.

Proof. Just direct matrix computation. □

Similarly, AE is the result of applying an elementary column operation. Note, however, that $A \mapsto AC_{i,j,\lambda}$ replaces $\mathbf{c}_j$ with $\mathbf{c}_j + \lambda \mathbf{c}_i$.

Also, elementary row operations preserve $\text{Row}(A)$ and elementary column operations preserve $\text{Col}(A)$. Thus, elementary row/column operations preserve $\text{r-rk}(A) = \text{c-rk}(A)$.

Definition 2.6 – Row-reduced echelon form

$A \in \mathcal{M}_{m \times n}(\mathbb{F})$ is in row-reduced echelon form if all following holds:

- (i) all the nonzero rows of A appear above all the zero rows.
- (ii) the leftmost nonzero element of a nonzero row is 1; we call this the pivot entry $1 \leq p(\mathbf{r}_i) \leq n$ of $\mathbf{r}_i$.
- (iii) if $\mathbf{r}_i, \mathbf{r}_j \neq \mathbf{0}$ and $i < j$, then $p(\mathbf{r}_i) < p(\mathbf{r}_j)$.
- (iv) in a column containing a pivot entry, every other entry is 0.

Example 2.1

$$\begin{pmatrix} 1 & a & 0 & 0 & b \\ 0 & 0 & 1 & 0 & c \\ 0 & 0 & 0 & 1 & d \end{pmatrix}$$

is in row-reduced echelon form. Similarly, A is in column-reduced echelon form if $A^\top$ is in row-reduced echelon form.

Lemma 2.3

If A is in row-reduced echelon form, then $\text{r-rk}(A) = \text{number of nonzero rows in } A$.

Proof. Let $\mathbf{r}_1, \dots, \mathbf{r}_k$ be the nonzero rows of A . Let $j_i = p(\mathbf{r}_i)$ be the pivot entry, so that $j_1 < \dots < j_k$. We claim that $\mathbf{r}_1, \dots, \mathbf{r}_k$ is a basis for $\text{Row}(A)$. Certainly, $\mathbf{r}_1, \dots, \mathbf{r}_k$ span $\text{Row}(A)$. Suppose $\mathbf{v} = \sum_{i=1}^k \lambda_i \mathbf{r}_i = \mathbf{0}$ with $\lambda_i \in \mathbb{F}$, then $\mathbf{0} = (\mathbf{v})_{j_i} = \lambda_i$. Hence, they are linearly independent. $\square$

Proposition 2.3

Every matrix $A \in \mathcal{M}_{m \times n}(\mathbb{F})$ can be put into row-reduced echelon form via elementary row operations.

Proof. We proceed by induction on n . Write $A = (\mathbf{c}_1 \mid \dots \mid \mathbf{c}_n)$. If $\mathbf{c}_1 = \mathbf{0}$, apply induction to $(\mathbf{c}_2 \mid \dots \mid \mathbf{c}_n)$. If $\mathbf{c}_1 \neq \mathbf{0}$, suppose the entry $(i, 1)$ is nonzero. Applying move (i), we can suppose the entry $(1, 1)$ is nonzero. Applying move (ii), we can assume the entry $(1, 1)$ is 1. Applying move (iii), we can purge all other entries in $\mathbf{c}_1$. We then obtain

$$\begin{pmatrix} 1 & * & \dots & * \\ 0 & & & \\ \vdots & & A' & \\ 0 & & & \end{pmatrix}.$$

By induction, we can use elementary row operations on rows 2 to m to reduce A' into row-reduced echelon form. Finally, use type (iii) moves to purge nonzero entries in row 1 and pivot columns. $\square$

- (i) Putting a matrix into row-reduced echelon form preserves the row-space.
- (ii) The row-reduced echelon form of any matrix is unique. The proof is omitted here.
- (iii) Transposing, we can put any matrix into column-reduced echelon form using elementary column operations.
- (iv) If A is a square matrix in row-reduced echelon form, then either A has a zero row or $A = I_m$.

Theorem 2.6

For $A \in \mathcal{M}_{n \times n}(\mathbb{F})$, the following are equivalent:

- (i) $\text{rk}(A) = n$;
- (ii) A is a product of elementary matrices;
- (iii) A is invertible.

Proof. For (i) implies (ii), by Proposition 2.3, there exist elementary matrices E_i such that $E_1 \dots E_l A$ is in row-reduced echelon form. By point (iv) above, this is equal to I_n so $A = E_l^{-1} \dots E_1^{-1}$.

For (ii) implies (iii), elementary matrices lie in $\text{GL}_n(\mathbb{F})$, which is a group.

For (iii) implies (i), suppose there exists $B \in \mathcal{M}_{n \times n}(\mathbb{F})$ such that $AB = I_n$. Then, for any $\mathbf{v} \in \mathbb{F}^n$, $\mathbf{v} = (AB)\mathbf{v} = A(B\mathbf{v})$ so $\mathbf{v} \in \text{im}(A)$. $\square$

With this, we have a constructive proof that A is equivalent to $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$.

Proof. We apply elementary matrices $E_1, \dots, E_l$ such that $E_1 \dots E_l A$ is in row-reduced echelon form. Apply elementary column operations of type (i) to move pivot-columns to the left. We obtain

$$E_1 \dots E_l A E'_1 \dots E'_k = \begin{bmatrix} I_r & * \\ 0 & 0 \end{bmatrix}.$$

Finally, we can use elementary column operations of type (iii) to purge $*$. We then obtain

$$PAQ = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$$

where $P \in \text{GL}_m(\mathbb{F})$ and $Q \in \text{GL}_n(\mathbb{F})$. $\square$

If $A \in \text{GL}_n(\mathbb{F})$, then we can reduce it to I_n using only elementary row operations or only elementary column operations.

3 Determinants and traces

For $A \in \mathcal{M}_{n \times n}(\mathbb{F})$, write $A = (\mathbf{c}_1 \mid \dots \mid \mathbf{c}_n)$ where $\mathbf{c}_i \in \mathbb{F}^n$ is the i -th column of A .

Theorem 3.1

There exists a unique function $F : \mathcal{M}_{n \times n}(\mathbb{F}) \rightarrow \mathbb{F}$ such that it is

- (i) alternating, if $\mathbf{c}_i = \mathbf{c}_j$ for some $i \neq j$, then $F(A) = 0$;
- (ii) multilinear in columns, for all $1 \leq i \leq n$ and $\mathbf{v}_j \in \mathbb{F}^n$, the function

$$\begin{aligned} \mathbb{F}^n &\rightarrow \mathbb{F} \\ \mathbf{v} &\mapsto F(\mathbf{v}_1 \mid \dots \mid \mathbf{v}_{i-1} \mid \mathbf{v} \mid \mathbf{v}_{i+1} \mid \dots \mid \mathbf{v}_n) \text{ is linear;} \end{aligned}$$

- (iii) non-trivial with $F(I_n) = 1$.

We shall call F the n -dimensional determinant. A function F that satisfies only (i) and (ii) is called a n -dimensional volume form.

Proof. To prove this, we would need the following lemma.

Lemma 3.1

Let F be a n -dimensional volume form and $A \in \mathcal{M}_{n \times n}(\mathbb{F})$, then

- (i) if A has a zero column, then $F(A) = 0$
- (ii) $F(AT_{i,j}) = -F(A)$
- (iii) $F(AM_{i,\lambda}) = \lambda F(A)$
- (iv) $F(AC_{i,j,\lambda}) = F(A)$

Proof. Let

$$\begin{aligned} f_i : \mathbb{F}^n &\rightarrow \mathbb{F} \\ \mathbf{v} &\mapsto F(\mathbf{c}_1 \mid \dots \mid \mathbf{c}_{i-1} \mid \mathbf{v} \mid \mathbf{c}_{i+1} \mid \dots \mid \mathbf{c}_n) \end{aligned}$$

so f_i is linear. Then, $f_i(\mathbf{c}_j) = \delta_{ij}F(A)$.

- (i) If $\mathbf{c}_i = \mathbf{0}$, then $F(A) = f_i(\mathbf{c}_i) = f_i(\mathbf{0}) = 0$.
- (ii) Let $\bar{A}$ be the matrix obtained from A by replacing both the i -th and j -th column of A by $\mathbf{c}_i + \mathbf{c}_j$. Then, $0 = F(\bar{A}) = F(A) + f_i(\mathbf{c}_j) + f_j(\mathbf{c}_i) + F(AT_{i,j})$. Since $f_i(\mathbf{c}_j) = f_j(\mathbf{c}_i) = 0$, we get the result.
- (iii) $F(AM_{i,\lambda}) = f_i(\lambda \mathbf{c}_i) = \lambda f_i(\mathbf{c}_i) = \lambda F(A)$.
- (iv) $F(AC_{i,j,\lambda}) = f_i(\mathbf{c}_i + \lambda \mathbf{c}_j) = f_i(\mathbf{c}_i) + \lambda f_i(\mathbf{c}_j) = F(A)$.

□

Now, we can prove the theorem. For uniqueness, let F be a n -dimensional volume form satisfying $F(I_n) = 1$. By the lemma, $F(T_{i,j}) = 1$, $F(M_{i,\lambda}) = \lambda$, $F(C_{i,j,\lambda}) = 1$ and $F(AE) = F(A)F(E)$ where E is elementary. Let $A \in \mathcal{M}_{n \times n}(\mathbb{F})$, we know there exist elementary matrices E_i such that $A' = AE_1 \dots E_l$ is in column-reduced echelon form. Then, $F(A) = F(A')F(E_1)^{-1} \dots F(E_l)^{-1}$. There are two cases.

- (i) A is invertible and $A' = I_n$, so $F(A) = F(E_1)^{-1} \dots F(E_l)^{-1}$.
- (ii) A' has a zero column, so by part (i) of the lemma, $F(A) = F(A') = 0$.

The proof of uniqueness yields the following.

Corollary 3.1

Let $A \in \mathcal{M}_{n \times n}(\mathbb{F})$, $\det(A) \neq 0$ if and only if A is invertible. In this case, $A = E_1 \dots E_l$ where E_i are elementary matrices, then $\det(A) = \det(E_1) \dots \det(E_l)$.

Before we prove existence, recall from IA Groups, S_n is the group of permutations of $\{1, \dots, n\}$. The sign homomorphism $\text{sgn} : S_n \rightarrow \{\pm 1\}$ is the unique homomorphism such that $\text{sgn}(\tau) = -1$ for all transpositions $\tau = (k \ l)$. $A_n = \ker(\text{sgn})$ so that $S_n = A_n \sqcup A_n\tau$ for all transpositions τ .

For existence, define $\det : \mathcal{M}_{n \times n}(\mathbb{F}) \rightarrow \mathbb{F}$ by

$$\det(A) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n a_{\sigma(i),i}.$$

We check that $\det$ satisfies all the conditions.

(i) Suppose $\mathbf{c}_k = \mathbf{c}_l$ with $k \neq l$. Set $\tau = (k \ l)$ so that $a_{i,j} = a_{i,\tau(j)}$ for all i, j . Then,

$$\begin{aligned} \det(A) &= \sum_{\sigma \in A_n} \operatorname{sgn}(\sigma) \prod_{i=1}^n a_{\sigma(i),i} + \sum_{\sigma \in A_n} \operatorname{sgn}(\sigma\tau) \prod_{i=1}^n a_{(\sigma\tau)(i),i} \\ &= \sum_{\sigma \in A_n} \prod_{i=1}^n a_{\sigma(i),i} - \sum_{\sigma \in A_n} \prod_{i=1}^n a_{(\sigma\tau)(i),i} \\ &= \sum_{\sigma \in A_n} \prod_{i=1}^n a_{\sigma(i),i} - \sum_{\sigma \in A_n} \prod_{i=1}^n a_{(\sigma\tau)(i),\tau(i)} \\ &= \sum_{\sigma \in A_n} \prod_{i=1}^n a_{\sigma(i),i} - \sum_{\sigma \in A_n} \prod_{j=1}^n a_{\sigma(j),j} = 0 \end{aligned}$$

since $\tau(i) = j$ ranges over $\{1, \dots, n\}$ just as i does.

(ii) Each product $\prod_{i=1}^n a_{\sigma(i),i}$ is multilinear in columns, hence so is $\det$.

(iii) $\det(I_n) = 1$ since only $\sigma = \text{id}$ contributes to the sum.

This concludes the proof. $\square$

Lemma 3.2

For $A \in \mathcal{M}_{n \times n}(\mathbb{F})$, $\det(A^\top) = \det(A)$.

Proof.

$$\begin{aligned} \det(A^\top) &= \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^n a_{i,\sigma(i)} \\ &= \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{j=1}^n a_{\sigma^{-1}(j),j} \\ &= \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma^{-1}) \prod_{j=1}^n a_{\sigma^{-1}(j),j} \\ &= \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{j=1}^n a_{\sigma(j),j} \\ &= \det(A) \end{aligned}$$

$\square$

Proposition 3.1

For all $A, B \in \mathcal{M}_{n \times n}(\mathbb{F})$, $\det(AB) = \det(A) \det(B)$.

Proof. By Lemma 2.1, $\operatorname{rk}(AB) \leq \min(\operatorname{rk}(A), \operatorname{rk}(B))$. If either A or B has $\operatorname{rk} < n$, then so does AB , hence $\det(AB) = 0 = \det(A) \det(B)$. Otherwise, we can write $A = E_1 \dots E_l$ and $B = E'_1 \dots E'_k$ with E_i, E'_j elementary matrices. Then, $AB = E_1 \dots E_l E'_1 \dots E'_k$ and $\det(AB) = \det(E_1) \dots \det(E_l) \det(E'_1) \dots \det(E'_k) = \det(E_1 \dots E_l) \det(E'_1 \dots E'_k)$. $\square$

Corollary 3.2

If A is invertible, then $\det(A^{-1}) = \det(A)^{-1}$.

Proof. $1 = \det(I_n) = \det(AA^{-1}) = \det(A) \det(A^{-1})$. $\square$

Corollary 3.3

Similar matrices have the same determinant.

Proof. Let A and A' be similar, then $A' = PAP^{-1}$ where $P \in \text{GL}_n(\mathbb{F})$. So, $\det(A') = \det(PAP^{-1}) = \det(P) \det(A) \det(P)^{-1} = \det(A)$. $\square$

Definition 3.1 – Determinant of a linear map

Let V be a finite-dimensional vector space with basis B and $\alpha \in \mathcal{L}(V, V)$. Define $\det(\alpha) = \det([\alpha]_B^B)$.

Proposition 3.2

$\det(\alpha)$ is independent of B .

Proof. Let B' be another basis for V and let $P = [\text{id}_V]_{B'}^B$. Recall that P is invertible with inverse $P^{-1} = [\text{id}_V]_B^{B'}$. Then, $[\alpha]_{B'}^{B'} = [\text{id}_V]_{B'}^B [\alpha]_B^B [\text{id}_V]_B^{B'} = P[\alpha]_B^B P^{-1}$ is similar to $[\alpha]_B^B$ and so they have the same determinant. $\square$

Definition 3.2 – Singular linear map

$\alpha \in \mathcal{L}(V, V)$ is called singular if $\det(\alpha) = 0$.

Corollary 3.4

For $\alpha \in \mathcal{L}(V, V)$, the following are equivalent:

- (i) $\text{rk}(\alpha) = \dim(V)$;
- (ii) α is invertible;
- (iii) α is non-singular.

Proof. (i) equivalent to (ii) is direct from the rank-nullity theorem.

For (i) equivalent to (iii), fix any basis B for V and set $A = [\alpha]_B^B$. Then, $\text{rk}(A) = \text{rk}(\alpha)$. Hence, $\text{rk}(\alpha) = \dim(V)$ if and only if $\text{rk}(A) = \dim(V)$ if and only if $\det(A) \neq 0$. $\square$

Proposition 3.3

Let $A \in \mathcal{M}_{k \times k}(\mathbb{F})$ and $B \in \mathcal{M}_{l \times l}(\mathbb{F})$. Let

$$C = \begin{bmatrix} A & * \\ 0 & B \end{bmatrix} \in \mathcal{M}_{n \times n}(\mathbb{F})$$

be a block triangular matrix. Then, $\det(C) = \det(A) \det(B)$.

Proof. We know $c_{ij} = 0$ for all $i \geq k+1$ and $j \leq k$. For $\sigma \in S_n$, if $\prod_{i=1}^n c_{\sigma(i),i} \neq 0$, then σ preserves $X = \{1, \dots, k\}$ and hence preserves $Y = \{k+1, \dots, n\}$. So, we can write $\sigma = \tau\theta$ where $\tau \in \text{Sym}(X)$

permutes X and fixes Y , and $\theta \in \text{Sym}(Y)$ permutes Y and fixes X . Conversely, given $\tau \in \text{Sym}(X)$ and $\theta \in \text{Sym}(Y)$, we can obtain $\sigma = \tau\theta \in S_n$ that preserves X and Y . Moreover,

$$\prod_{i=1}^n c_{(\tau\theta)(i),i} = \left(\prod_{x \in X} c_{(\tau\theta)(x),x} \right) \left(\prod_{y \in Y} c_{(\tau\theta)(y),y} \right) = \left(\prod_{x \in X} c_{\tau(x),x} \right) \left(\prod_{y \in Y} c_{\theta(y),y} \right).$$

Thus,

$$\begin{aligned} \det(C) &= \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n c_{\sigma(i),i} \\ &= \sum_{\substack{\tau \in \text{Sym}(X) \\ \theta \in \text{Sym}(Y)}} \text{sgn}(\tau\theta) \prod_{i=1}^n c_{(\tau\theta)(i),i} \\ &= \sum_{\tau, \theta} \text{sgn}(\tau\theta) \left(\prod_{x \in X} c_{\tau(x),x} \right) \left(\prod_{y \in Y} c_{\theta(y),y} \right) \\ &= \sum_{\tau} \sum_{\theta} \text{sgn}(\tau) \text{sgn}(\theta) \left(\prod_x c_{\tau(x),x} \right) \left(\prod_y c_{\theta(y),y} \right) \\ &= \left(\sum_{\tau} \text{sgn}(\tau) \prod_x c_{\tau(i),i} \right) \left(\sum_{\theta} \text{sgn}(\theta) \prod_y c_{\theta(i),i} \right) \\ &= \det(A) \det(B). \end{aligned}$$

□

Generalising, for

$$A = \begin{bmatrix} A_1 & * & \dots & * \\ 0 & A_2 & & \\ \vdots & & \ddots & \\ 0 & & & A_k \end{bmatrix}$$

a block triangular matrix where A_i are square matrices, $\det(A) = \prod_{i=1}^k \det(A_i)$.

Definition 3.3 – Minor

For $A \in \mathcal{M}_{n \times n}(\mathbb{F})$, the (i, j) -minor matrix of A is $A_{i\hat{j}} \in \mathcal{M}_{(n-1) \times (n-1)}(\mathbb{F})$, obtained from A by deleting the i -th row and j -th column of A .

Lemma 3.3 – Laplace expansion along a column or row

- (i) For any fixed $1 \leq i \leq n$, $\det(A) = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det(A_{i\hat{j}})$.
- (ii) For any fixed $1 \leq j \leq n$, $\det(A) = \sum_{i=1}^n (-1)^{i+j} a_{ij} \det(A_{i\hat{j}})$.

Proof. (ii) follows from (i) by transposing and using $\det(C^\top) = \det(C)$. Recall that the elementary matrix $T_{i,j}$ that swaps two rows or columns has determinant -1 . Let $A = (\mathbf{c}_1 \mid \dots \mid \mathbf{c}_n)$ so $\mathbf{c}_j = \sum_{i=1}^n a_{ij} \mathbf{e}_i$. By linearity in column j ,

$$\det(A) = \sum_{i=1}^n a_{ij} \det(\underbrace{(\mathbf{c}_1 \mid \dots \mid \mathbf{c}_{j-1} \mid \mathbf{e}_i \mid \mathbf{c}_{j+1} \mid \dots \mid \mathbf{c}_n)}_{\text{call this } A^{(i)}}).$$

It suffices to find $\det(A^{(i)})$. Applying $(j-1)$ elementary column operations $T_{k,k+1}$, we obtain $\det(A^{(i)}) =$

$(-1)^{j-1} \det((\mathbf{e}_1 | \mathbf{c}_1 | \dots | \mathbf{c}_{j-1} | \mathbf{c}_{j+1} | \dots | \mathbf{c}_n))$. Applying $(i-1)$ elementary row operations $T_{k,k+1}$, we get

$$\det(A^{(i)}) = (-1)^{i-1} (-1)^{j-1} \det \begin{pmatrix} 1 & * & \dots & * \\ 0 & & & \\ \vdots & & A_{\widehat{i},\widehat{j}} & \\ 0 & & & \end{pmatrix}.$$

This is a block triangular matrix, so $\det(A^{(i)}) = (-1)^{i+j} \det(A_{\widehat{i},\widehat{j}})$. □

Definition 3.4 – Adjugate

The adjugate matrix $\text{adj}(A)$ of A is given by $\text{adj}(A) = ((-1)^{i+j} \det(A_{\widehat{j},\widehat{i}}))_{i,j} \in \mathcal{M}_{n \times n}(\mathbb{F})$.

Theorem 3.2 – Cramer's rule

- (i) $\text{adj}(A)A = A \text{adj}(A) = \det(A)I_n$.
- (ii) If A is invertible, then $A^{-1} = \det(A)^{-1} \text{adj}(A)$.

Proof. (ii) is immediate from (i) since inverses are unique. For (i), by the Laplace expansion on the j -th column,

$$\det(A) = \sum_i (\text{adj}(A))_{j,i} a_{ij} = (j, j)\text{-entry of } \text{adj}(A)A.$$

For $j \neq k$, replace the j -th column of A by the k -th column,

$$0 = \det((\mathbf{c}_1 | \dots | \mathbf{c}_k | \dots | \mathbf{c}_k | \dots | \mathbf{c}_n)) = \sum_i (\text{adj}(A))_{j,i} a_{ik} = (j, k)\text{-entry of } \text{adj}(A)A.$$

$A \text{adj}(A)$ is the same, just expanding on a row. □

Definition 3.5 – Trace

For $A \in \mathcal{M}_{n \times n}(\mathbb{F})$, the trace of A is given by

$$\text{tr}(A) = \sum_{i=1}^n a_{ii}.$$

We see that $\text{tr} \in \mathcal{L}(\mathcal{M}_{n \times n}(\mathbb{F}), \mathbb{F})$.

Lemma 3.4

For all $A, B \in \mathcal{M}_{n \times n}(\mathbb{F})$, $\text{tr}(AB) = \text{tr}(BA)$.

Proof. $\text{tr}(AB) = \sum_{i=1}^n (AB)_{ii} = \sum_{i=1}^n \sum_{j=1}^n a_{ij} b_{ji} = \sum_{j=1}^n \sum_{i=1}^n b_{ji} a_{ij} = \sum_{j=1}^n (BA)_{jj} = \text{tr}(BA)$ □

Unlike determinants, $\text{tr}(AB) \neq \text{tr}(A)\text{tr}(B)$ in general.

Corollary 3.5

Similar matrices have the same trace.

Proof. For $P \in \text{GL}_n(\mathbb{F})$, $\text{tr}(PAP^{-1}) = \text{tr}(P^{-1}(PA)) = \text{tr}(IA) = \text{tr}(A)$. □

Definition 3.6 – Trace of a linear map

Let V be a finite-dimensional vector space and $\alpha \in \mathcal{L}(V, V)$, we can define the trace of α by $\text{tr}(\alpha) = \text{tr}([\alpha]_B^B)$ where B is a basis for V .

Proposition 3.4

This is independent of the basis B .

Proof. If B' is another basis for V , then $[\alpha]_{B'}^{B'}$ and $[\alpha]_B^B$ are similar. □

4 Dual spaces

Definition 4.1 – Dual space

Let V be a $\mathbb{F}$ -vector space. The dual space of V is

$$V^* = \mathcal{L}(V, \mathbb{F}) = \{\theta : V \rightarrow \mathbb{F} \mid \theta \text{ is linear}\}.$$

Example 4.1

- (i) $\theta : \mathbb{R}^3 \rightarrow \mathbb{R}$ given by $\theta((x \ y \ z)^\top) = x - 2y + 3z$.
- (ii) $\text{tr} \in \mathcal{M}_{n \times n}(\mathbb{F})^*$.
- (iii) Let $V = C([0, 1])$ and $\theta : V \rightarrow \mathbb{R}$ be given by $\theta(f) = \int_0^1 f(t)e^{-t} dt$. Then, $\theta \in V^*$.

An element of θ of V^* is called a linear functional on V .

Proposition 4.1

Suppose B is a basis for V . For $b \in B$, define $b^* \in V^*$ by

$$b^* \left(\sum'_{c \in B} \lambda_c c \right) = \lambda_b,$$

that is $b^*(c) = \delta_{b,c}$ for $c \in B$. Set $B^* = \{b^* : b \in B\} \subseteq V^*$. Then,

- (i) B^* is linearly independent.
- (ii) if V is finite-dimensional, then B^* is a basis for V^* .

Definition 4.2 – Dual basis

If V is finite-dimensional, we call B^* the dual basis to B .

Proof.

- (i) Suppose $\sum'_{b \in B} \lambda_b b^* = \underline{0}$ in V^* . Then, for $c \in B$, $0 = (\sum'_{b \in B} \lambda_b b^*)(c) = \sum'_{b \in B} \lambda_b b^*(c) = \lambda_c$.

- (ii) We give two proofs. We know that if V and W are finite-dimensional, then $\dim(\mathcal{L}(V, W)) = \dim(V)\dim(W)$, so $\dim(V^*) = \dim(\mathcal{L}(V, \mathbb{F})) = \dim(V)$. B^* is a linearly independent subset of V^* of size $\dim(V) = \dim(V^*)$, hence B^* is a basis for V^* .

Now, a constructive proof. Given $\theta \in V^*$ and $b \in B$, set $\lambda_b = \theta(b) \in \mathbb{F}$ and take $\bar{\theta} = \sum'_{b \in B} \lambda_b b^* \in V^*$. Then, $\bar{\theta} \in \langle B^* \rangle$ and for $c \in B$, $\bar{\theta}(c) = \lambda_c = \theta(c)$. So, $\bar{\theta} = \theta$ as they agree on a basis for V , and hence $\theta \in \langle B^* \rangle$.

□

Example 4.2

V being finite-dimensional is required for (ii). Take $V = P(\mathbb{R}) =$ set of all real polynomial functions. Then, by example sheet 2, $V^* \cong \mathbb{R}^{\mathbb{N}}$. But, by example sheet 1, $\mathbb{R}^{\mathbb{N}} \not\cong P(\mathbb{R})$.

Corollary 4.1

For V being finite-dimensional, $V \cong V^*$.

Proof. By Proposition 4.1, $\dim(V) = \dim(V^*)$.

□

Let V be a vector space with $\dim(V) = n$ and a basis $B = \{v_1, \dots, v_n\}$. For $v \in V$ and $\theta \in V^*$, write $v = \sum_{i=1}^n \lambda_i v_i$ and $\theta = \sum_{j=1}^n \mu_j v_j^*$. Then, $\theta(v) = \sum_{i,j} \lambda_i \mu_j v_j^*(v_i) = \sum_{i=1}^n \lambda_i \mu_i = ([\theta]_{B^*})^\top \cdot [v]_B$, which is the dot product between $([\theta]_{B^*})^\top$ and $[v]_B$.

Definition 4.3 – Annihilator

Let V be a $\mathbb{F}$ -vector space and $S \subseteq V$. The annihilator of S is

$$S^0 = \{\theta \in V^* : \theta(s) = 0 \text{ for all } s \in S\} \subseteq V^*.$$

Lemma 4.1

For $S, T \subseteq V$, then

- (i) $S^0 \leq V^*$
- (ii) if $S \subseteq T$, then $T^0 \leq S^0$
- (iii) $S^0 = \langle S \rangle^0$
- (iv) $V^0 = \{0_{V^*}\}$ and $\{0_V\}^0 = V^*$

Proof.

(i) Apply the subspace test.

(ii) By (i), it suffices to check $T^0 \subseteq S^0$. For $\theta \in T^0$ and $s \in S$, we have $s \in T$ so $\theta(s) = 0$.

(iii) $S \subseteq \langle S \rangle$ so $\langle S \rangle^0 \leq S^0$ by (ii). Conversely, let $\theta \in S^0$ and $v \in \langle S \rangle$, write $v = \sum'_{s \in S} \lambda_s s$ so $\theta(v) = \sum'_{s \in S} \lambda_s \theta(s) = 0$ and hence $\theta \in \langle S \rangle^0$.

(iv) If $\theta \in V^*$ and for all $v \in V$, $\theta(v) = 0$, then θ is the zero functional. For $\theta \in V^*$, $\theta(0_V) = 0$ so $\theta \in \{0_V\}^0$.

□

Proposition 4.2

Let V be a finite-dimensional vector space and $U \leq V$, then

$$\dim(V) = \dim(U) + \dim(U^0).$$

Proof. Let $\dim(V) = n$ and $\dim(U) = k$. Let $B_U = \{v_1, \dots, v_k\}$ be a basis for U and extend it to a basis $B_V = \{v_1, \dots, v_n\}$ for V . Then, $B_V^* = \{v_1^*, \dots, v_n^*\}$ is a basis for V^* . So, it suffices to prove that $\{v_{k+1}^*, \dots, v_n^*\}$ is a basis for U^0 and we need to check the following three properties.

It is a subset of U^0 , for $i \leq k$ and $j \geq k+1$, $v_j^*(v_i) = 0$ so $v_j^* \in (B_U)^0 = \langle B_U \rangle^0 = U^0$.

It is linearly independent as a subset of the linearly independent set B_V^* .

It spans, let $\theta \in U^0$ and write $\theta = \sum_{j=1}^n \lambda_j v_j^*$. Then, for $i \leq k$, $v_i \in U$ so $0 = \theta(v_i) = \sum_j \lambda_j v_j^*(v_i) = \lambda_i$. Hence, $\theta = \sum_{j=k+1}^n \lambda_j v_j^* \in \langle v_{k+1}^*, \dots, v_n^* \rangle$. $\square$

More generally, in any vector space V , if $U, W \leq V$ satisfy $V = U \oplus W$, then $U^0 \cong W^*$.

Proposition 4.3

Let V be a $\mathbb{F}$ -vector space and $U, W \leq V$, then

- (i) $U^0 \cap W^0 = (U + W)^0$;
- (ii) $U^0 + W^0 \leq (U \cap W)^0$;
- (iii) if V is finite-dimensional, we have equality in (ii).

Proof.

(i)

$$\begin{aligned} \theta \in (U + W)^0 &\iff \theta(u + w) = 0 \text{ for all } u \in U \text{ and } w \in W \\ &\iff \theta(u) = \theta(w) = 0 \text{ for all } u \in U \text{ and } w \in W \\ &\iff \theta \in U^0 \cap W^0. \end{aligned}$$

(ii) $U \cap W \leq U, W$. By part (ii) of Lemma 4.1, $U^0, W^0 \leq (U \cap W)^0$ so $U^0 + W^0 \leq (U \cap W)^0$.

(iii) Let $n = \dim(V)$, then

$$\begin{aligned} \dim(U^0 + W^0) &= \dim(U^0) + \dim(W^0) - \dim(U^0 \cap W^0) \\ &= \dim(U^0) + \dim(W^0) - \dim((U + W)^0) \\ &= (n - \dim(U)) + (n - \dim(W)) - (n - \dim(U + W)) \\ &= n - \dim(U \cap W) \\ &= \dim((U \cap W)^0). \end{aligned}$$

$\square$

Definition 4.4 – Dual map

Let $\alpha \in \mathcal{L}(V, W)$. The dual map of α is $\alpha^* : W^* \rightarrow V^*$ given by $\alpha^*(\theta) = \theta \circ \alpha$.

Lemma 4.2

If $\alpha, \beta \in \mathcal{L}(V, W)$ and $\gamma \in \mathcal{L}(U, V)$ and $\lambda \in \mathbb{F}$, then

- (i) α^* is linear;
- (ii) $(\alpha + \lambda\beta)^* = \alpha^* + \lambda\beta^*$;
- (iii) $(\alpha \circ \gamma)^* = \gamma^* \circ \alpha^*$;
- (iv) if β is an isomorphism, then so is β^* with $(\beta^*)^{-1} = (\beta^{-1})^*$.

Proof.

- (i) Let $\theta, \eta \in W^*$ and $\mu \in \mathbb{F}$. Then, for $v \in V$,

$$\begin{aligned} (\alpha^*(\theta + \mu\eta))(v) &= (\theta + \mu\eta)(\alpha(v)) \\ &= \theta(\alpha(v)) + \mu\eta(\alpha(v)) \\ &= (\alpha^*(\theta))(v) + \mu(\alpha^*(\eta))(v) \\ &= (\alpha^*(\theta) + \mu\alpha^*(\eta))(v). \end{aligned}$$

This is true for all v so $\alpha^*(\theta + \mu\eta) = \alpha^*(\theta) + \mu\alpha^*(\eta)$.

- (ii) For $\theta \in W^*$ and $v \in V$,

$$\begin{aligned} ((\alpha + \lambda\beta)^*(\theta))(v) &= \theta((\alpha + \lambda\beta)(v)) \\ &= \theta(\alpha(v) + \lambda\beta(v)) \\ &= \theta(\alpha(v)) + \lambda\theta(\beta(v)) \\ &= (\alpha^*(\theta))(v) + \lambda(\beta^*(\theta))(v) \\ &= (\alpha^*(\theta) + \lambda\beta^*(\theta))(v). \end{aligned}$$

This is true for all v so $(\alpha + \lambda\beta)^*(\theta) = \alpha^*(\theta) + \lambda\beta^*(\theta) = (\alpha^* + \lambda\beta^*)(\theta)$.

- (iii) For $\theta \in W^*$,

$$\begin{aligned} (\alpha \circ \gamma)^*(\theta) &= \theta \circ (\alpha \circ \gamma) \\ &= (\theta \circ \alpha) \circ \gamma \\ &= \alpha^*(\theta) \circ \gamma \\ &= \gamma^*(\alpha^*(\theta)) \\ &= (\gamma^* \circ \alpha^*)(\theta). \end{aligned}$$

This is true for all θ so $(\alpha \circ \gamma)^* = \gamma^* \circ \alpha^*$.

- (iv) Note that for all $\theta \in V^*$, $(\text{id}_V)^*(\theta) = \theta \circ \text{id}_V = \theta$ so $(\text{id}_V)^* = \text{id}_{V^*}$. Thus, $\text{id}_{V^*} = (\beta^{-1} \circ \beta)^* = \beta^* \circ (\beta^{-1})^*$. Similarly, $\text{id}_{W^*} = (\beta^{-1})^* \circ \beta^*$.

□

Proposition 4.4

Let V, W be finite-dimensional vector spaces and $\alpha \in \mathcal{L}(V, W)$. Let B, C be bases for V, W respectively. Then,

$$[\alpha^*]_{B^*}^{C^*} = ([\alpha]_C^B)^\top$$

Proof. Let $n = \dim(V)$ and $m = \dim(W)$. Let $B = \{b_1, \dots, b_n\}$ and $C = \{c_1, \dots, c_m\}$. Let $A = [\alpha]_C^B$, so

$\alpha(b_i) = \sum_{k=1}^m a_{ki}c_k$. Let $A' = [\alpha^*]_{B^*}^{C^*}$, so $\alpha^*(c_j^*) = \sum_{l=1}^n a'_{lj}b_l^*$. Then,

$$a'_{ij} = (\alpha^*(c_j^*))(b_i) = c_j^*(\alpha(b_i)) = c_j^*\left(\sum_{k=1}^m a_{ki}c_k\right) = a_{ji}.$$

□

This is useful when changing basis. Let B, C be bases for a finite-dimensional V . Set $P = [\text{id}_V]_C^B$, then $[\text{id}_{V^*}]_{B^*}^{C^*} = P^\top$ so $[\text{id}_{V^*}]_{C^*}^{B^*} = (P^\top)^{-1}$.

Corollary 4.2

Let V, W be finite-dimensional vector spaces and $\alpha \in \mathcal{L}(V, W)$. Then, $\text{rk}(\alpha^*) = \text{rk}(\alpha)$.

Proof. Set $A = [\alpha]_C^B$. Recall that $\text{rk}(\alpha) = \text{c-rk}(A)$. But, $\text{c-rk}(A) = \text{r-rk}(A) = \text{c-rk}(A^\top) = \text{rk}(\alpha^*)$. □

Proposition 4.5

Let V, W be $\mathbb{F}$ -vector spaces, for $\alpha \in \mathcal{L}(V, W)$,

- (i) $\ker(\alpha^*) = \text{im}(\alpha)^0$;
- (ii) $\text{im}(\alpha^*) \leq \ker(\alpha)^0$;
- (iii) if V, W are finite-dimensional, we have equality for (ii).

Proof.

- (i) For $\theta \in W^*$,

$$\begin{aligned} \theta \in \ker(\alpha^*) &\iff \theta \circ \alpha = \underline{0} \text{ in } V^* \\ &\iff \theta(\alpha(v)) = 0 \text{ for all } v \in V \\ &\iff \theta \in \text{im}(\alpha)^0. \end{aligned}$$

- (ii) For $\eta \in V^*$,

$$\begin{aligned} \eta \in \text{im}(\alpha^*) &\implies \text{there exists } \theta \in W^* \text{ such that } \eta = \theta \circ \alpha \\ &\implies \text{for all } v \in \ker(\alpha), \eta(v) = \theta(\alpha(v)) = \theta(\underline{0}) = 0 \\ &\implies \eta \in \ker(\alpha)^0. \end{aligned}$$

- (iii) By Corollary 4.2, $\text{rk}(\alpha^*) = \text{rk}(\alpha) = \dim(V) - \dim(\ker(\alpha)) = \dim(\ker(\alpha)^0)$.

□

Definition 4.5 – Double dual

The double dual of V is $V^{**} = (V^*)^*$.

If V is finite-dimensional, then $V^{**} \cong V$ since they have the same dimension.

Theorem 4.1

Let V be a $\mathbb{F}$ -vector space, there is a linear map $\varepsilon : V \rightarrow V^{**}$ given by $(\varepsilon(v))(\theta) = \theta(v)$ for $v \in V$ and $\theta \in V^*$. If V is finite-dimensional, then ε is an isomorphism.

Proof. For linearity, let $v, w \in V$ and $\lambda \in \mathbb{F}$ and $\theta \in V^*$.

$$\begin{aligned} (\varepsilon(v + \lambda w))(\theta) &= \theta(v + \lambda w) \\ &= \theta(v) + \lambda\theta(w) \\ &= (\varepsilon(v))(\theta) + \lambda(\varepsilon(w))(\theta) \\ &= (\varepsilon(v) + \lambda\varepsilon(w))(\theta). \end{aligned}$$

This is true for all θ so $\varepsilon(v + \lambda w) = \varepsilon(v) + \lambda\varepsilon(w)$.

Now suppose V is finite-dimensional. First, we prove injectivity by showing that the kernel is trivial. Suppose there is $v \in V$ with $v \neq 0$ and $\varepsilon(v) = 0$. That is, for all $\theta \in V^*$, $(\varepsilon(v))(\theta) = \theta(v) = 0$. Extend a basis for V by $\{v_1 = v, v_2, \dots, v_n\}$. Let $B^* = \{v_1^*, \dots, v_n^*\}$ be a dual basis. Set $\theta = v_1^*$, then $0 = \theta(v) = v_1^*(v_1) = 1$ and contradiction. Surjectivity follows from the rank-nullity theorem since $\dim(V) = \dim(V^{**})$. $\square$

We call $\varepsilon : V \rightarrow V^{**}$ the evaluation map, or the natural isomorphism. It is natural since it does not involve a choice of basis. For a finite-dimensional V , we also have $V \cong V^*$, but any isomorphism requires a choice of basis.

Let V, W be finite-dimensional vector spaces. Let B, C be bases for V, W respectively and let $\alpha \in \mathcal{L}(V, W)$. Then,

$$\begin{aligned} (\alpha^*(\theta))(v) &= ([\alpha^*(\theta)]_{B^*})^\top \cdot [v]_B \\ &= (A^\top [\theta]_{C^*})^\top \cdot [v]_B \text{ where } A = [\alpha]_C^B \\ &= ([\theta]_{C^*})^\top \cdot (A[v]_B) \\ &= ([\theta]_{C^*})^\top \cdot [\alpha(v)]_C. \end{aligned}$$

Proposition 4.6

Let V be a finite-dimensional vector space. Every basis C for V^* is the dual basis to some basis of V .

Proof. Let $C = \{\theta_1, \dots, \theta_n\}$. Let $C^* = \{\theta_1^*, \dots, \theta_n^*\} \subset V^{**}$ be the dual basis. Let $\varepsilon : V \rightarrow V^{**}$ be the natural isomorphism. Set $v_i = \varepsilon^{-1}(\theta_i^*) \in V$. Then, $B = \{v_1, \dots, v_n\}$ is a basis of V since it is the image of the basis C^* under the isomorphism ε^{-1} . Then, for all i and j , $\theta_i(v_j) = (\varepsilon(v_j))(\theta_i) = \theta_j^*(\theta_i) = \delta_{ij}$. Hence, C is the dual basis to B . $\square$

Proposition 4.7

Let V be finite-dimensional and $U \leq V$, $\varepsilon(U) = (U^0)^0 \leq V^{**}$.

Proof. For $u \in U$ and $\theta \in U^0$, $(\varepsilon(u))(\theta) = \theta(u) = 0$. This is true for all θ so $\varepsilon(U) \leq (U^0)^0$. But

$$\dim((U^0)^0) = \dim(V^*) - \dim(U^0) = \dim(V) - (\dim(V) - \dim(U)) = \dim(\varepsilon(U)).$$

Hence, $\varepsilon(U) = (U^0)^0$. $\square$

It is common to identify V with V^{**} under ε . Under this identification, the previous proposition becomes $U = (U^0)^0$. For $X \subset V^*$, $X^0 \leq V^{**}$ is identified with

$$X_0 = \{v \in V : \theta(v) = 0 \text{ for all } \theta \in X\} = \bigcap_{\theta \in X} \ker(\theta) \leq V.$$

Then, $\dim(X_0) = n - \dim(\langle X \rangle)$. Also, every subspace $U \leq V$ is X_0 for some $X \in V^*$, namely $X = U^0$.

Before we start the next chapter, let us reintroduce polynomials.

Definition 4.6 – Polynomial

A polynomial f over a field $\mathbb{F}$ is a formal expression

$$f(t) = \sum_{i=0}^n a_i t^i$$

where $n \in \mathbb{Z}_{\geq 0}$ and $a_i \in \mathbb{F}$ are the coefficients.

$\mathbb{F}[t]$ is the $\mathbb{F}$ -vector space of all polynomials. The degree $\deg(f)$ of $f \in \mathbb{F}[t]$ is the largest i such that $a_i \neq 0$, and we take $\deg(0) = -\infty$. This a_i is called the leading coefficient, and f is monic if $a_i = 1$.

We can do arithmetic on polynomials. Let $f(t) = \sum_{i=0}^n a_i t^i$ and $g(t) = \sum_{j=0}^m b_j t^j$. $(f + g)$ has t^i -coefficient $(a_i + b_i)$. (fg) has t^i -coefficient $\sum_{k=0}^i a_k b_{i-k}$. Note that $\deg(f + g) \leq \max\{\deg(f), \deg(g)\}$ and $\deg(fg) = \deg(f) + \deg(g)$. We write $f \mid g$ if there exists $h \in \mathbb{F}[t]$ such that $g = fh$. For $\lambda \in \mathbb{F}$, write $f(\lambda) = \sum_{i=0}^n a_i \lambda^i \in \mathbb{F}$, so that $(f + g)(\lambda) = f(\lambda) + g(\lambda)$ and $(fg)(\lambda) = f(\lambda)g(\lambda)$.

There is a difference between $\mathbb{F}[t]$ and the space of polynomial maps $\mathbb{F} \rightarrow \mathbb{F}$. If $\mathbb{F}$ is finite, then $\mathbb{F}[t]$ is not finite-dimensional, but $\{\text{polynomial maps on } \mathbb{F}\}$ is a subspace of $\mathbb{F}^{\mathbb{F}}$ and hence is finite-dimensional.

Proposition 4.8 – Euclid’s algorithm for polynomials

For $f, g \in \mathbb{F}[t]$ with $g \neq 0$, there exist $q, r \in \mathbb{F}[t]$ such that $f = qg + r$ with $\deg(r) < \deg(g)$.

Proof. Let $n = \deg(f)$ and $m = \deg(g) \geq 0$. If $m > n$, set $q = 0$ and $r = f$. If $n \geq m$, set $f'(t) = f(t) - (a_n/b_m)t^{n-m}g(t)$ so that $\deg(f') < \deg(f)$. Do induction and we are done. $\square$

Corollary 4.3 – Bezout’s lemma

If $f_1, \dots, f_n \in \mathbb{F}[t]$ has no common divisor of degree ≥ 1 , then there exist $q_1, \dots, q_n \in \mathbb{F}[t]$ such that $1 = \sum_{i=1}^n f_i q_i$ in $\mathbb{F}[t]$.

Proof. The same as for integers, using Euclid’s algorithm. $\square$

Lemma 4.3

For $\lambda \in \mathbb{F}$, $f(\lambda) = 0$ if and only if $(t - \lambda) \mid f(t)$.

Proof. Apply Euclid’s algorithm to $f(t)$ and $g(t) = t - \lambda$. $\square$

Definition 4.7 – Root and multiplicity

$\lambda \in \mathbb{F}$ is a root of $f \in \mathbb{F}[t]$ of multiplicity $\geq e$ if $(t - \lambda)^e \mid f(t)$.

Corollary 4.4

If $\deg(f) = n \geq 0$, then f has at most n roots, counted with multiplicity.

Corollary 4.5

If $\deg(f), \deg(g) < n$ and there exist distinct $\lambda_1, \dots, \lambda_n \in \mathbb{F}$ such that $f(\lambda_i) = g(\lambda_i)$ for all i , then $f = g$.

Theorem 4.2 – Fundamental theorem of algebra

Every $f \in \mathbb{C}[t]$ of degree $n \geq 1$ has exactly n roots, counted with multiplicity.

5 Eigenspaces

Definition 5.1 – Diagonalisable and triangularisable

Let V be a finite-dimensional vector space and $\alpha \in \mathcal{L}(V, V)$. α is diagonalisable (and respectively triangularisable) if there exists a basis B of V such that $[\alpha]_B^B$ is a diagonal matrix (respectively an upper-triangular matrix).

$A \in \mathcal{M}_{n \times n}(\mathbb{F})$ is diagonalisable (and respectively triangularisable) if A is similar to a diagonal matrix (respectively an upper-triangular matrix).

By a change of basis, α is diagonalisable (and respectively triangularisable) if and only if for any basis B of V , $[\alpha]_B^B$ is a diagonalisable (respectively triangularisable). Why do we care? If A is triangularisable, then it is easy to compute $\det(A)$. If A is diagonalisable, it is easier to understand the similarity class of A .

If $B = \{v_1, \dots, v_n\}$ is such that $[\alpha]_B^B = \text{diag}(\lambda_1, \dots, \lambda_n)$, then $\alpha(v_i) = \lambda_i v_i$ for all i . This prompts the following definition.

Definition 5.2 – Eigenvalue, eigenvector and eigenspace

Let V be a $\mathbb{F}$ -vector space and $\alpha \in \mathcal{L}(V, V)$. $\lambda \in \mathbb{F}$ is an eigenvalue of α if there exists $0 \neq v \in V$ such that $\alpha(v) = \lambda v$. Such a vector v is a λ -eigenvector of α . $V_\lambda = \{v \in V : \alpha(v) = \lambda v\}$ is the λ -eigenspace of α .

(i) $V_\lambda = \ker(\alpha - \lambda \text{id}_V) \leq V$.

(ii) For a finite-dimensional V , α is diagonalisable if and only if V has a basis of eigenvectors of α .

(iii) If $v \in V_\mu$ for $\lambda \neq \mu$, then $(\alpha - \lambda \text{id}_V)(v) = (\mu - \lambda)(v)$. Thus, $(\alpha - \lambda \text{id}_V)$ preserves V_μ and $(\alpha - \lambda \text{id}_V)|_{V_\mu}$ is invertible with inverse $v \mapsto (\mu - \lambda)^{-1}v$.

Lemma 5.1

If $\lambda_1, \dots, \lambda_k \in \mathbb{F}$ are distinct and $0 \neq v_i \in V_{\lambda_i}$, then $\{v_1, \dots, v_k\}$ is linearly independent.

Proof. Suppose they are not linearly independent. Let $\sum_{i=1}^k \mu_i v_i = 0$ in V and without loss of generality, set $\mu_1 \neq 0$. Let $\beta = \prod_{i=2}^k (\alpha - \lambda_i \text{id}_V)$. Then, by (iii) above,

$$\beta(v_j) = \left(\prod_{i=2}^k (\lambda_j - \lambda_i) \right) v_j \neq 0$$

if and only if $j = 1$. Thus,

$$0 = \beta \left(\sum_{i=1}^k \mu_i v_i \right) = \left(\mu_1 \prod_{i=2}^k (\lambda_1 - \lambda_i) \right) v_1$$

so $\mu_1 = 0$, hence contradiction. $\square$

Corollary 5.1

If V is finite-dimensional, then every $\alpha \in \mathcal{L}(V, V)$ has only finitely many eigenvalues.

Proposition 5.1

If V is finite-dimensional and $\alpha \in \mathcal{L}(V, V)$. Let $\lambda_1, \dots, \lambda_k \in \mathbb{F}$ be all the eigenvalues of α . Then,

- (i) $\langle V_{\lambda_1} \cup \dots \cup V_{\lambda_k} \rangle = V_{\lambda_1} \oplus \dots \oplus V_{\lambda_k}$.
- (ii) α is diagonalisable if and only if $V = V_{\lambda_1} \oplus \dots \oplus V_{\lambda_k}$.

Proof.

- (i) Suppose that it is not true. Then, there exists $v_j \in V_{\lambda_j}$ not all 0 such that $\sum_j v_j = 0$. This is a contradiction by the Lemma 5.1.
- (ii) If α is diagonalisable, by (ii) of the remark, V has a basis of eigenvectors, so the V_{λ_i} span V . Conversely, let B_i be a basis of V_{λ_i} . Then, by question 8 of example sheet 1, $\bigcup_{i=1}^n B_i = B$ is a basis for V . Hence, V has a basis of eigenvectors. $\square$

Recall that for V finite-dimensional and $\beta \in \mathcal{L}(V, V)$, $\det(\beta) = 0$ if and only if $\ker(\beta) \neq \{0_V\}$. Thus, $\lambda \in \mathbb{F}$ is an eigenvalue of $\alpha \in \mathcal{L}(V, V)$ if and only if $\det(\alpha - \lambda \text{id}_V) = 0$.

Definition 5.3 – Characteristic polynomial

For $A \in \mathcal{M}_{n \times n}(\mathbb{F})$, the characteristic polynomial of A is $\chi_A(t) = \det(tI_n - A) \in \mathbb{F}[t]$.

Similarly, for $\alpha \in \mathcal{L}(V, V)$ with V finite-dimensional, $\chi_\alpha(t) = \det(t \text{id}_V - \alpha)$.

Lemma 5.2

$\{\text{eigenvalues of } \alpha\} = \{\text{roots of } \chi_\alpha\}$.

- (i) χ_α is monic of degree $\dim(V)$.
- (ii) Similar matrices have the characteristic polynomial.
- (iii) By the Leibniz formula, if $\chi_\alpha(t) = t^n + C_{n-1}t^{n-1} + \dots + C_1t + C_0$, then $C_{n-1} = -\text{tr}(\alpha)$ and $C_0 = (-1)^n \det(\alpha)$.

Proposition 5.2

If α is triangularisable, then $\chi_\alpha(t)$ can be written as a product of linear factors.

Proof. Let B be a basis of V such that

$$A = [\alpha]_B^B = \begin{pmatrix} a_1 & & * \\ & \ddots & \\ 0 & & a_n \end{pmatrix}.$$

Then,

$$\chi_\alpha(t) = \det \begin{pmatrix} t - a_1 & & * \\ & \ddots & \\ 0 & & t - a_n \end{pmatrix} = \prod_{i=1}^n (t - a_i).$$

□

Example 5.1

Consider a rotation by angle θ in $\mathbb{R}^2$.

$$A = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

Then, $\chi_A(t) = t^2 - (2 \cos \theta)t + 1$ is irreducible for $\theta \in (0, \pi)$. Hence, A is not triangularisable in $\mathcal{M}_{2 \times 2}(\mathbb{R})$.

Theorem 5.1

Every $A \in \mathcal{M}_{n \times n}(\mathbb{C})$ is triangularisable.

Proof. By the fundamental theorem of algebra, $\chi_A(t)$ has a root $\lambda \in \mathbb{C}$, so there exists $\underline{0} \neq v_1 \in \mathbb{C}^n$ such that $Av_1 = \lambda v_1$. Extend $\{v_1\}$ to a basis $B = \{v_1, \dots, v_n\}$ of $\mathbb{C}^n$. So, up to a change of basis, we can assume

$$A = \begin{pmatrix} \lambda & * & \dots & * \\ 0 & & & \\ \vdots & & C & \\ 0 & & & \end{pmatrix}.$$

By induction, there exists $\bar{P} \in \text{GL}_{n-1}(\mathbb{C})$ such that $\bar{P}C\bar{P}^{-1}$ is upper triangular. Set

$$P = \begin{pmatrix} 1 & * & \dots & * \\ 0 & & & \\ \vdots & & \bar{P} & \\ 0 & & & \end{pmatrix} \in \text{GL}_n(\mathbb{C}),$$

then PAP^{-1} is upper triangular. □

The same proof says that for any field $\mathbb{F}$, $A \in \mathcal{M}_{n \times n}(\mathbb{F})$ is triangularisable if and only if $\chi_A(t)$ is a product of linear factors. This is because, in the proof, $\chi_A(t) = (t - \lambda)\chi_C(t)$.

Let $\alpha \in \mathcal{L}(V, V)$. For $p(t) = \sum_{i=0}^n \mu_i t^i$, we can write $p(\alpha) = \sum_{i=0}^n \mu_i \alpha^i \in \mathcal{L}(V, V)$ where $\alpha^0 = \text{id}_V$. Similarly, for $A \in \mathcal{M}_{k \times k}(\mathbb{F})$, write $p(A) = \sum_{i=0}^n \mu_i A^i$.

(i) If B is a basis for V , then $[p(\alpha)]_B^B = p([\alpha]_B^B)$.

(ii) If $\dim(V) = n$, then $\dim(\mathcal{L}(V, V)) = n^2$. So, $\{\text{id}_V, \alpha, \alpha^2, \dots, \alpha^{n^2}\}$ is linearly dependent. Thus, there exist $\mu_i \in \mathbb{F}$ not all zero such that $0 = \sum_{i=0}^{n^2} \mu_i \alpha^i = p(\alpha)$ where $p(t) = \sum_{i=0}^{n^2} \mu_i t^i \neq 0$.

Let $I_\alpha = \{p(t) \in \mathbb{F}[t] : p(\alpha) = 0\}$. Then, $I_\alpha \neq \{0\}$ by (ii) of the remark. Let $m_\alpha \in I_\alpha$ be a nonzero element of minimal degree. Rescaling, we can assume m_α is monic.

Definition 5.4 – Minimal polynomial

Let $I_\alpha = \{p(t) \in \mathbb{F}[t] : p(\alpha) = 0\}$. Let $m_\alpha \in I_\alpha$ be the nonzero monic element of minimal degree. m_α is the minimal polynomial of α .

Proposition 5.3

The minimal polynomial m_α is unique.

Proof. To prove this, we need the following lemma.

Lemma 5.3

For any $f \in I_\alpha$, we have $m_\alpha \mid f$.

Proof. By the Euclidean algorithm, there exist $r, q \in \mathbb{F}[t]$ such that $\deg(r) < \deg(m_\alpha)$ and $f = qm_\alpha + r$. Then, $r(\alpha) = f(\alpha) - q(\alpha)m_\alpha(\alpha) = 0$ since $f(\alpha) = m_\alpha(\alpha) = 0$. So, $r \in I_\alpha$ but by the minimality of $\deg(m_\alpha)$, we have $r = 0$. $\square$

Let $\bar{m}_\alpha$ be another nonzero monic element of minimal degree in I_α . By Lemma 5.3, $m_\alpha \mid \bar{m}_\alpha$ but $\deg(m_\alpha) = \deg(\bar{m}_\alpha)$, so there exists $\lambda \in \mathbb{F}$ such that $\bar{m}_\alpha = \lambda m_\alpha$. Since m_α and $\bar{m}_\alpha$ are both monic, so $\lambda = 1$. $\square$

Theorem 5.2 – Cayley-Hamilton

Let χ_α be the characteristic polynomial of $\alpha \in \mathcal{L}(V, V)$ where V is finite-dimensional, then $\chi_\alpha(\alpha) = 0$.

Proof. Here, we only give a proof valid for $\mathbb{F} = \mathbb{C}$. A proof for arbitrary $\mathbb{F}$ is on example sheet 3.

By theorem 5.1, there exists a basis $B = \{v_1, \dots, v_n\}$ for V such that

$$[\alpha]_B^B = \begin{pmatrix} \lambda_1 & & * \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}.$$

Then, $\chi_\alpha(t) = \prod_{i=1}^n (t - \lambda_i)$. Set $U_j = \langle v_1, \dots, v_j \rangle$ (so that $U_0 = \{0_V\}$ and $U_n = V$). Then, for $1 \leq j \leq n$, $(\alpha - \lambda_j \text{id}_V)U_j \leq U_{j-1}$. Thus, $(\chi_\alpha(\alpha))(V) = ((\alpha - \lambda_1 \text{id}_V) \circ \dots \circ (\alpha - \lambda_n \text{id}_V))(U_n) = U_0 = \{0_V\}$ so $\chi_\alpha(\alpha) = 0$. $\square$

Corollary 5.2

$m_\alpha \mid \chi_\alpha$

Proof. By the Cayley-Hamilton theorem and Lemma 5.3. $\square$

Definition 5.5 – Algebraic and geometric multiplicity

Let $\lambda \in \mathbb{F}$ be an eigenvalue of $\alpha \in \mathcal{L}(V, V)$.

- (i) The algebraic multiplicity a_λ of λ is the multiplicity of λ as a root of $\chi_\alpha(t)$.
- (ii) The geometric multiplicity g_λ of λ is $\dim(V_\lambda)$.

Proposition 5.4 – The other AM-GM inequality

$$g_\lambda \leq a_\lambda$$

Proof. Let $\{v_1, \dots, v_k\}$ be a basis for V_λ . Extend it to a basis $B = \{v_1, \dots, v_n\}$ for V . Then, $[\alpha]_B^B = \begin{bmatrix} \lambda I_k & * \\ 0 & C \end{bmatrix}$. Thus, $\chi_\alpha(t) = (t - \lambda)^k \chi_C(t)$ so $a_\lambda \geq k = g_\lambda$. $\square$

Lemma 5.4

Let the eigenvalues of α be $\lambda_1, \dots, \lambda_k \in \mathbb{F}$. Then,

- (i) $\sum_{i=1}^k g_{\lambda_i} \leq \dim(V)$ with equality if and only if α is diagonalisable.
- (ii) $\sum_{i=1}^k a_{\lambda_i} \leq \dim(V)$ with equality if and only if α is triangularisable.

Proof.

- (i) By Proposition 5.1, $V_{\lambda_1} \oplus \dots \oplus V_{\lambda_k} \leq V$ with equality if and only if α is diagonalisable.
- (ii) Write $\chi_\alpha(t) = f(t) \prod_{i=1}^k (t - \lambda_i)^{a_{\lambda_i}}$, then $f(t)$ has no linear factors since a root of χ_α is an eigenvalue. Moreover, $\dim(V) = \deg(\chi_\alpha) = \deg(f) + \sum_{i=1}^k a_{\lambda_i} \geq \sum_{i=1}^k a_{\lambda_i}$ with equality if and only if χ_α is a product of linear factors, which only happens if α is triangularisable by Proposition 5.2. $\square$

Proposition 5.5

For $\mathbb{F} = \mathbb{C}$, α is diagonalisable if and only if $a_\lambda = g_\lambda$ for every eigenvalue λ of α .

Proof. By Theorem 5.1, α is triangularisable. So, by Lemma 5.4, $\sum_{i=1}^k a_{\lambda_i} = \dim(V)$. By Proposition 5.4, $\sum_{i=1}^k g_{\lambda_i} \leq \sum_{i=1}^k a_{\lambda_i} = \dim(V)$ with equality if and only if $a_{\lambda_i} = g_{\lambda_i}$ for all i . This occurs if and only if α is diagonalisable by Lemma 5.4. $\square$

For $p(t) = \sum_{i=1}^n \mu_i t^i$ and $v \in V_\lambda$, we have $(p(\alpha))(v) = \sum_{i=1}^n \mu_i \alpha^i(v) = \sum_{i=1}^n \mu_i \lambda^i v = p(\lambda)v$.

Lemma 5.5

For $\lambda \in \mathbb{F}$, λ is a root of m_α if and only if λ is a root of χ_α .

Proof. If λ is a root of m_α , by the Cayley-Hamilton theorem, $(t - \lambda) \mid m_\alpha(t) \mid \chi_\alpha(t)$.

If $\chi_\alpha(\lambda) = 0$, then λ is an eigenvalue of α , so let $0 \neq v \in V_\lambda$. Then, $(m_\alpha(\alpha))(v) = m_\alpha(\lambda)v$. But $m_\alpha(\alpha) = 0$ and $v \neq 0$, so $m_\alpha(\lambda) = 0$. $\square$

Let c_λ be the multiplicity of λ as a root of m_α . By Cayley-Hamilton, if λ is an eigenvalue, then $1 \leq c_\lambda \leq a_\lambda$.

There is no relationship between c_λ and g_λ .

Example 5.2

(i) Let $A = \lambda I_n$. Then, $\chi_A(t) = (t - \lambda)^n$ and $V_\lambda \in \mathbb{F}^n$, so $m_\alpha(t) = t - \lambda$. Hence, $a_\lambda = n$, $g_\lambda = n$ and $c_\lambda = 1$.

(ii) Let

$$A = \begin{pmatrix} \lambda & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda \end{pmatrix}.$$

Then, $\chi_A(t) = (t - \lambda)^n$ and $V_\lambda = \langle \mathbf{e}_1 \rangle$, so $a_\lambda = n$ and $g_\lambda = 1$.

$$(A - \lambda I_n)^{n-1} = \begin{pmatrix} 0 & \dots & 0 & 1 \\ \vdots & & & 0 \\ \vdots & & & \vdots \\ 0 & \dots & \dots & 0 \end{pmatrix} \neq 0$$

so $C_\lambda = n$ as $m_\alpha = \chi_\alpha$.

Theorem 5.3

Let $\alpha \in \mathcal{L}(V, V)$ with V being finite-dimensional. If there exists $0 \neq p(t) \in \mathbb{F}[t]$ such that p is a product of distinct linear factors and $p(\alpha) = 0$, then α is diagonalisable.

Proof. We can assume p is monic. Write $p(t) = \prod_{j=1}^k (t - \mu_j)$ with $\mu_j \in \mathbb{F}$ distinct. Set $p_i(t) = \prod_{j \neq i} (t - \mu_j)$. Then, by Bezout's lemma, there exist $q_i(t) \in \mathbb{F}[t]$ such that $1 = \sum_{i=1}^k p_i(t) q_i(t)$ in $\mathbb{F}[t]$. Set $\pi_i = p_i(\alpha) \circ q_i(\alpha) \in \mathcal{L}(V, V)$. Then, $\text{id}_V = \sum_{i=1}^k \pi_i$, so for any $v \in V$, $v = \text{id}_V(v) = \sum_{i=1}^k \pi_i(v)$. Thus, $V = \text{im}(\pi_1) + \dots + \text{im}(\pi_k)$. Moreover, $(t - \mu_i)p_i(t) = p(t)$ so $(\alpha - \mu_i \text{id}_V) \circ p_i(\alpha) = p(\alpha) = 0$. Thus, $(\alpha - \mu_i \text{id}_V) \circ \pi_i = 0$ so $\text{im}(\pi_i) \leq V_{\mu_i}$. Hence, $V = \sum_i \text{im}(\pi_i) \leq \bigoplus_i V_{\mu_i} \leq V$ so $V = \bigoplus_i V_{\mu_i}$ and α is diagonalisable. $\square$

Example 5.3

If $A \in \text{GL}_n(\mathbb{C})$ has finite order, then A is diagonalisable. Let $m = \text{ord}(A)$, that is $A^m = I_n$, so $p(A) = 0$ where $p(t) = t^m - 1 = \prod_{i=1}^m (t - \omega^i)$ and $\omega = e^{2\pi i/m}$.

In summary,

Theorem 5.4

The following are equivalent:

- (i) α is diagonalisable;
- (ii) V has a basis, consisting of the eigenvectors of α ;
- (iii) there exists $0 \neq p(t) \in \mathbb{F}[t]$ which is a product of distinct linear factors such that $p(\alpha) = 0$;
- (iv) $m_\alpha(t)$ is a product of distinct linear factors.

Moreover, if $\mathbb{F} = \mathbb{C}$, these are equivalent to (v) $a_\lambda = g_\lambda$ for all $\lambda \in \mathbb{C}$.

Proof. (i) is equivalent to (ii) by Proposition 5.1. (i) is equivalent to (v) by Proposition 5.5. (iii) is equivalent to (iv) by Lemma 5.3. (iii) implies (i) by Theorem 5.3. For (i) implies (iii), let B be a basis for V such that $[\alpha]_B^B$ is diagonal. Rearranging B , we can assume

$$[\alpha]_B^B = \begin{bmatrix} \lambda_1 I_{n_1} & & 0 \\ & \ddots & \\ 0 & & \lambda_n I_{n_n} \end{bmatrix}$$

with λ_i distinct. Set $p(t) = \prod_{i=1}^k (t - \lambda_i)$. Then, $[p(\alpha)]_B^B = p([\alpha]_B^B) = 0$, hence $p(\alpha) = 0$. $\square$

Definition 5.6 – Simultaneously diagonalisable

$\alpha, \beta \in \mathcal{L}(V, V)$ are simultaneously diagonalisable if there exists basis B for V such that $[\alpha]_B^B$ and $[\beta]_B^B$ are diagonal.

Theorem 5.5

Suppose α and β are diagonalisable. Then, they are simultaneously diagonalisable if and only if they commute, that is $\alpha \circ \beta = \beta \circ \alpha$.

Proof. For the $\implies$ direction, note that diagonal matrices commute. Given B as in Definition 5.6, we have $[\alpha \circ \beta]_B^B = [\alpha]_B^B [\beta]_B^B = [\beta]_B^B [\alpha]_B^B = [\beta \circ \alpha]_B^B$.

For the $\impliedby$ direction, we need

Lemma 5.6

If α and β commute and V_λ is the λ -eigenspace of α , then $\beta(V_\lambda) \leq V_\lambda$.

Proof. For $v \in V_\lambda$, $\alpha(\beta(v)) = \beta(\alpha(v)) = \beta(\lambda v) = \lambda(\beta v)$ so $\beta(v) \in V_\lambda$. $\square$

α is diagonalisable so we can let $V = V_{\lambda_1} \oplus \cdots \oplus V_{\lambda_k}$ where V_{λ_i} is the λ_i -eigenspace of A and λ_i are distinct. Let $0 \neq p(t) \in \mathbb{F}[t]$ be a product of distinct linear factors such that $p(\beta) = 0$. By Lemma 5.6, we can define $\beta|_{V_{\lambda_i}} : V_{\lambda_i} \rightarrow V_{\lambda_i}$. Then, $p(\beta|_{V_{\lambda_i}}) = p(\beta)|_{V_{\lambda_i}} = 0$, hence $\beta|_{V_{\lambda_i}}$ is diagonalisable. Let B_i be a basis of V_{λ_i} , consisting of eigenvectors of β . Then, $B = \bigcup_{i=1}^k B_i$ is a basis for V , and consists of vectors which are eigenvectors for both α and β . Thus, $[\alpha]_B^B$ and $[\beta]_B^B$ are both diagonal. $\square$

Definition 5.7 – Jordan matrix

For $\lambda \in \mathbb{C}$, the $(n \times n)$ -Jordan matrix is

$$J_n(\lambda) = \begin{pmatrix} \lambda & 1 & & 0 \\ & \ddots & \ddots & \\ 0 & & \ddots & 1 \\ & & & \lambda \end{pmatrix} \in \mathcal{M}_{n \times n}(\mathbb{C}).$$

By Example 5.2, $\chi_{J_n(\lambda)}(t) = (t - \lambda)^n = m_{J_n(\lambda)}(t)$. Also, the λ -eigenspace is $\langle \mathbf{e}_1 \rangle$.

Definition 5.8 – Jordan normal form

$A \in \mathcal{M}_{n \times n}(\mathbb{C})$ is in Jordan normal form if

$$A = \begin{bmatrix} J_{n_1}(\lambda_1) & & 0 \\ & \ddots & \\ 0 & & J_{n_k}(\lambda_k) \end{bmatrix}$$

for some $n_i \in \mathbb{N}$ such that $\sum_{i=1}^k n_i = n$ and some $\lambda_i \in \mathbb{C}$ not necessarily distinct. The $J_{n_i}(\lambda_i)$ are the Jordan blocks of A .

Theorem 5.6

Every $A \in \mathcal{M}_{n \times n}(\mathbb{C})$ is similar to a matrix $\text{JNF}(A)$ in Jordan normal form. $\text{JNF}(A)$ is unique up to rearrangement of the Jordan blocks.

Proof. First, we prove uniqueness. For $A, \bar{A} \in \mathcal{M}_{n \times n}(\mathbb{C})$ with $A \sim \bar{A}$ and $\bar{A}$ in Jordan normal form, define $R_{\lambda,r}(\bar{A})$ to be the number of λ -Jordan-blocks in $\bar{A}$ of size $\geq r$ for $\lambda \in \mathbb{C}$ and $r \in \mathbb{N}$. Then, $R_{\lambda,r}(\bar{A})$ determine $\bar{A}$ uniquely up to rearrangement of the Jordan blocks.

Lemma 5.7

$$R_{\lambda,r}(\bar{A}) = \text{rk}((A - \lambda I_n)^{r-1}) - \text{rk}((A - \lambda I_n)^r).$$

Proof. For $P \in \text{GL}_n(\mathbb{C})$, $P(A - \lambda I_n)^k P^{-1} = (P(A - \lambda I_n)P^{-1})^k = (PAP^{-1} - \lambda I_n)^k$. Thus, $\text{rk}((\bar{A} - \lambda I_n)^k) = \text{rk}((A - \lambda I_n)^k)$. Hence, we can assume $A = \bar{A}$ is in Jordan normal form.

$$J_l(\lambda) - \lambda I_l = \begin{pmatrix} 0 & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & 0 \end{pmatrix} \text{ and } (J_l(\lambda) - \lambda I_l)^2 = \begin{pmatrix} 0 & 0 & 1 & & 0 \\ & \ddots & \ddots & \ddots & \\ & & \ddots & \ddots & 1 \\ & & & \ddots & 0 \\ 0 & & & & 0 \end{pmatrix}$$

and so on. Thus,

$$\text{rk}((J_l(\lambda) - \lambda I_l)^k) = \begin{cases} l - k & \text{if } k \leq l \\ 0 & \text{if } k \geq l. \end{cases}$$

Also, for $\mu \neq \lambda$, $J_l(\mu) - \lambda I_l$ is invertible. Hence,

$$\begin{aligned} \text{rk}((A - \lambda I_n)^k) &= n - k(\text{number of } \lambda\text{-Jordan-blocks of } A \text{ of size } \geq k) \\ &\quad - \sum (\text{sizes of } \lambda\text{-Jordan-blocks of size } < k) \end{aligned}$$

which gives the result. $\square$

The result for uniqueness follows since the right hand side depends only on A .

The proof for existence is non-examinable and we will only provide a sketch here. A commutes with $A - \lambda_i I_{n_i}$ so A preserves V_i by an adaptation of Lemma 5.6, where V_i is the generalised λ_i -eigenspace. Thus,

$$A \sim \begin{bmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_k \end{bmatrix}$$

where $m_{A_i}(t) = (t - \lambda_i)^{c_i}$. It suffices to show that A_i have Jordan normal form since if $A_i \sim \bar{A}_i$, then

$$\begin{bmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_k \end{bmatrix} \sim \begin{bmatrix} \bar{A}_1 & & 0 \\ & \ddots & \\ 0 & & \bar{A}_k \end{bmatrix}.$$

Replacing A by A_i , then by $A_i - \lambda_i I$, we can assume that $A^m = 0$ for some $m \in \mathbb{N}$. That is, A is nilpotent.

Theorem 5.7

Let V be a n -dimensional vector space and $\alpha \in \mathcal{L}(V, V)$ be nilpotent. Then, there exists a basis $B = \{v_1, \dots, v_n\}$ for V such that $\alpha(v_i) \in \{v_{i-1}, 0\}$ for all $1 \leq i \leq n$.

Proof. We proceed by induction on $\dim(V)$. The base case is easy. Let $W = \text{im}(\alpha)$, then $\dim(W) < \dim(V)$ and $\alpha(W) \leq W$. So, we can define $\alpha|_W : W \rightarrow W$ which is nilpotent. By induction, we have a basis $B' = \{\alpha^{l_1}(w_1), \alpha^{l_1-1}(w_1), \dots, \alpha(w_1), w_1, \alpha^{l_2}(w_2), \dots, \alpha(w_2), w_2, \dots, \alpha^{l_k}(w_k), \dots, w_k\}$ for W with $\alpha^{l_i+1}(w_i) = 0$. Write $w_i = \alpha(v_i)$ for some $v_i \in V$ and extend $\{\alpha^{l_1}(w_1), \dots, \alpha^{l_k}(w_k)\} \subset \ker(\alpha)$ by $\{u_1, \dots, u_m\}$ to a basis for $\ker(\alpha)$. We check that $B = B' \cup \{u_1, \dots, u_m\} \cup \{v_1, \dots, v_k\}$ are linearly independent. By the rank-nullity theorem, $|B| = \dim(V)$. Up to reordering, B is as desired. $\square$

The result then follows. $\square$

Example 5.4

Let $\lambda_i \in \mathbb{C}$ be distinct. For $n = 2$, the possible Jordan normal forms are

$$A = \begin{pmatrix} \lambda_1 & 1 \\ 0 & \lambda_1 \end{pmatrix} \quad \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \quad \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_1 \end{pmatrix}$$

with $m_A(t) = \quad (t - \lambda_1)^2 \quad (t - \lambda_1)(t - \lambda_2) \quad (t - \lambda_1).$

For $n = 3$, we have

$$\begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix} \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_1 & 0 \\ 0 & 0 & \lambda_2 \end{pmatrix} \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_1 & 0 \\ 0 & 0 & \lambda_1 \end{pmatrix} \begin{pmatrix} \lambda_1 & 1 & 0 \\ 0 & \lambda_1 & 0 \\ 0 & 0 & \lambda_2 \end{pmatrix} \begin{pmatrix} \lambda_1 & 1 & 0 \\ 0 & \lambda_1 & 0 \\ 0 & 0 & \lambda_1 \end{pmatrix} \begin{pmatrix} \lambda_1 & 1 & 0 \\ 0 & \lambda_1 & 1 \\ 0 & 0 & \lambda_1 \end{pmatrix}.$$

These all have different minimal polynomials.

For $n \leq 3$, JNF(A) is uniquely determined by m_A and χ_A . This is false for $n \geq 4$.

Let

$$A = \begin{bmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_k \end{bmatrix} \in \mathcal{M}_{n \times n}(\mathbb{C})$$

with $A_i \in \mathcal{M}_{n_i \times n_i}(\mathbb{C})$.

(i) For $p(t) \in \mathbb{C}[t]$, we have

$$p(A) = \begin{bmatrix} p(A_1) & & 0 \\ & \ddots & \\ 0 & & p(A_k) \end{bmatrix}$$

so $m_A(t) = \text{lcm}(m_{A_1}(t), \dots, m_{A_k}(t))$. Hence, $c_\lambda(A) = \max_i c_\lambda(A_i)$.

(ii) $\chi_A(t) = \prod_{i=1}^k \chi_{A_i}(t)$ so $a_\lambda(A) = \sum_{i=1}^k a_\lambda(A_i)$.

(iii) $\text{rk}(A) = \sum_{i=1}^k \text{rk}(A_i)$ and

$$A - \lambda I_n = \begin{bmatrix} A_1 - \lambda I_{n_1} & & 0 \\ & \ddots & \\ 0 & & A_k - \lambda I_{n_k} \end{bmatrix}.$$

Hence, $g_\lambda(A) = n - \text{rk}(A - \lambda I_n) = \sum_{i=1}^k (n_i - \text{rk}(A_i - \lambda I_{n_i})) = \sum_{i=1}^k g_\lambda(A_i)$.

Proposition 5.6

For $A \in \mathcal{M}_{n \times n}(\mathbb{C})$ in Jordan normal form,

- (i) $a_\lambda(A)$ = sum of sizes of λ -Jordan-blocks in A .
- (ii) $g_\lambda(A)$ = number of λ -Jordan-blocks in A .
- (iii) $c_\lambda(A)$ = size of largest λ -Jordan-block in A .

Proof. Immediate from the above calculations and $a_\lambda(J_n(\lambda)) = c_\lambda(J_n(\lambda)) = n$ and $g_\lambda(J_n(\lambda)) = 1$. $\square$

Example 5.5

Consider

$$A = \begin{pmatrix} 3 & -2 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}.$$

We want to find $P \in \text{GL}_3(\mathbb{C})$ such that PAP^{-1} is in Jordan normal form. $\chi_A(t) = (t-1)^2(t-2)$ so the Jordan normal form is either

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \text{ or } \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

But

$$A - I_3 = \begin{pmatrix} 2 & -2 & 0 \\ 1 & -1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

has rank 2 and $\ker(A - I_3) = \langle (0 \ 0 \ 1)^\top \rangle$ so the latter is the actual Jordan normal form. Set $v_1 = (0 \ 0 \ 1)^\top$. Similarly, $\ker(A - 2I_3) = \langle v_3 \rangle$ where $v_3 = (2 \ 1 \ 2)^\top$. Finally, solve $Av_2 = v_1 + v_2$ for v_2 and get that $v_2 = (1 \ 1 \ 0)^\top$. Hence, $P = (v_1 \mid v_2 \mid v_3)$.

Definition 5.9 – Generalised eigenspace

Let V be a finite-dimensional $\mathbb{C}$ -vector space and $\alpha \in \mathcal{L}(V, V)$. Write $m_\alpha(t) = (t - \lambda_1)^{c_1} \dots (t - \lambda_k)^{c_k}$ with $\lambda_i \in \mathbb{C}$ distinct and $c_i \geq 1$. The generalised λ_i -eigenspace of α is

$$V_i = \ker((\alpha - \lambda_i \text{id}_V)^{c_i}) \leq V.$$

Theorem 5.8 – Generalised eigenspace decomposition

$V = V_1 \oplus \dots \oplus V_k$ where V_i is the generalised λ_i -eigenspace.

Proof. On example sheet 3. Adapt the proof of Theorem 5.3. $\square$

Lemma 5.8

If $\underline{0} \neq v_i \in V_i$ for each i , then $\{v_1, \dots, v_k\}$ are linearly independent.

Proof. Let $0 \leq d_i < c_i$ be such that $\underline{0} \neq w_i = (\alpha - \lambda_i \text{id}_V)^{d_i}(v_i)$ but $(\alpha - \lambda_i \text{id}_V)^{d_i+1}(v_i) = \underline{0}$. Then w_i is a λ_i -eigenvector of α so $\{w_1, \dots, w_k\}$ are linearly independent. Set $\beta = \prod_{j=1}^k (\alpha - \lambda_j \text{id}_V)^{d_j} \in \mathcal{L}(V, V)$. As the $(\alpha - \lambda_i \text{id}_V)$ commute, this is well-defined. Also,

$$\beta(v_i) = \left(\prod_{j \neq i} (\alpha - \lambda_j \text{id}_V)^{d_j} \right) (w_i) = \underbrace{\left(\prod_{j \neq i} (\lambda_i - \lambda_j)^{d_j} \right)}_{\text{call this } \mu_i \neq 0} w_i.$$

Suppose $\underline{0} = \sum_{i=1}^k v_i v_i$ with $v_i \in \mathbb{C}$. Then, $\underline{0} = \beta(\sum_i v_i v_i) = \sum_i \mu_i v_i w_i$ so $v_i = 0$. □

6 Bilinear forms

Definition 6.1 – Bilinear form

Let U and V be $\mathbb{F}$ -vector spaces. A bilinear form is a function $\varphi : U \times V \rightarrow \mathbb{F}$ such that for each fixed $u_0 \in U$ and $v_0 \in V$, both

$$\begin{array}{ccc} V & \rightarrow & \mathbb{F} \\ v & \mapsto & \varphi(u_0, v) \end{array} \quad \text{and} \quad \begin{array}{ccc} U & \rightarrow & \mathbb{F} \\ u & \mapsto & \varphi(u, v_0) \end{array}$$

are linear.

Example 6.1

- (i) Let $U = V = \mathbb{F}^n$, the dot product $\varphi(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^n x_i y_i$ is a bilinear form.
- (ii) For $A \in \mathcal{M}_{m \times n}(\mathbb{F})$, the map $\varphi_A : \mathbb{F}^m \times \mathbb{F}^n \rightarrow \mathbb{F}$ is bilinear.

$$(\mathbf{x}, \mathbf{y}) \mapsto \mathbf{x}^\top A \mathbf{y}$$
- (iii) Let $U = V = C([0, 1])$, $\varphi(f, g) = \int_0^1 f(t)g(t) dt$ is bilinear.
- (iv) The map $\varphi : V \times V^* \rightarrow \mathbb{F}$ is bilinear.

$$(v, \theta) \mapsto \theta(v)$$

Definition 6.2

Suppose U and V are finite-dimensional, $B = \{b_1, \dots, b_m\}$ and $C = \{c_1, \dots, c_n\}$ are bases for U and V respectively, and $\varphi : U \times V \rightarrow \mathbb{F}$ is bilinear. Then, the matrix of φ with respect to B and C is

$$[\varphi]_{B,C} = (\varphi(b_i, c_j))_{ij} \in \mathcal{M}_{m \times n}(\mathbb{F}).$$

Proposition 6.1

$[\varphi]_{B,C}$ satisfies

$$([u]_B)^\top [\varphi]_{B,C} [v]_C = \varphi(u, v)$$

for all $u \in U$ and $v \in V$. Moreover, $[\varphi]_{B,C}$ is the unique matrix with this property.

Proof. Let $u = \sum_{i=1}^m \lambda_i b_i$ and $v = \sum_{j=1}^n \mu_j c_j$ so that $[u]_B = \lambda$ and $[v]_C = \mu$. Then,

$$\varphi(u, v) = \sum_{i=1}^m \sum_{j=1}^n \lambda_i \mu_j \varphi(b_i, c_j) = \lambda^\top (\varphi(b_i, c_j))_{ij} \mu.$$

If $A \in \mathcal{M}_{m \times n}(\mathbb{F})$ is another matrix that satisfies this property, $\varphi(b_i, c_j) = ([b_i]_B)^\top A [c_j]_C = \mathbf{e}_i^\top A \mathbf{e}_j = a_{ij}$ so it is unique. $\square$

Corollary 6.1

If B, B' are bases for U and C, C' are bases for V , then

$$[\varphi]_{B', C'} = ([\text{id}_U]_B^{B'})^\top [\varphi]_{B, C} [\text{id}_V]_C^{C'}.$$

Proof. For $u \in U$ and $v \in V$,

$$\begin{aligned} ([u]_{B'})^\top ([\text{id}_U]_B^{B'})^\top [\varphi]_{B, C} [\text{id}_V]_C^{C'} [v]_{C'} &= ([\text{id}_U]_B^{B'} [u]_{B'})^\top [\varphi]_{B, C} [v]_C \\ &= ([u]_B)^\top [\varphi]_{B, C} [v]_C \\ &= \varphi(u, v) \end{aligned}$$

which gives the result by Proposition 6.1. $\square$

Definition 6.3 – Rank of a bilinear form

The rank $\text{rk}(\varphi)$ of φ is the rank of $[\varphi]_{B, C}$ for any bases B and C for U and V respectively.

- (i) Equivalent matrices have same rank, so by Corollary 6.1, $\text{rk}(\varphi)$ is well-defined.
- (ii) For $U = V$ and B, B' being bases for V , there exists $P \in \text{GL}_n(\mathbb{F})$ such that $[\varphi]_{B', B'} = P^\top [\varphi]_{B, B} P$.

Definition 6.4 – Congruent matrices

$A, A' \in \mathcal{M}_{n \times n}(\mathbb{F})$ are congruent if there exists $P \in \text{GL}_n(\mathbb{F})$ such that $A' = P^\top A P$.

We can check that congruence is an equivalence relation.

Definition 6.5

Let $\varphi : U \times V \rightarrow \mathbb{F}$ be a bilinear form. For $u \in U$ and $v \in V$, let $\varphi_L(u) \in V^*$ and $\varphi_R(v) \in U^*$ be given by

$$(\varphi_L(u))(v) = \varphi(u, v) = (\varphi_R(v))(u).$$

Lemma 6.1

$\varphi_L : U \rightarrow V^*$ and $\varphi_R : V \rightarrow U^*$ are linear.

Proof. As an exercise. $\square$

Definition 6.6 – Left-kernel and right-kernel

$\ker(\varphi_L) \leq U$ is the left-kernel of φ . $\ker(\varphi_R) \leq V$ is the right-kernel of φ .

Proposition 6.2

Suppose U, V are finite-dimensional and B, C are bases for U, V respectively. Let B^*, C^* be the dual bases to B, C respectively. Then,

$$[\varphi_L]_{C^*}^B = [\varphi]_{B,C} \text{ and } [\varphi_R]_{B^*}^C = ([\varphi]_{B,C})^\top.$$

Proof. Let $B = \{b_1, \dots, b_m\}$ and $C = \{c_1, \dots, c_n\}$. Set $\varphi_L(b_i) = \sum_{j=1}^n a'_{ij} c_j^*$. Write $[\varphi]_{B,C} = A$. Then,

$$a'_{ij} = (\varphi_L(b_i))(c_j) = \varphi(b_i, c_j) = a_{ij}.$$

Similarly for φ_R . □

Corollary 6.2

$$\text{rk}(\varphi_L) = \text{rk}(\varphi) = \text{rk}(\varphi_R).$$

Proof. Direct from Proposition 6.2. □

Definition 6.7

For $S \subseteq U$ and $T \subseteq V$, define

$$S^\perp = \{v \in V : \varphi(s, v) = 0 \text{ for all } s \in S\} \text{ and } {}^\perp T = \{u \in U : \varphi(u, t) = 0 \text{ for all } t \in T\}.$$

- (i) $S^\perp \leq V$ and ${}^\perp T \leq U$ by the subspace test.
- (ii) If $S_1 \subseteq S_2 \subseteq U$, then $S_2^\perp \leq S_1^\perp \leq V$. Similarly, if $T_1 \subseteq T_2 \subseteq V$, then ${}^\perp T_2 \leq {}^\perp T_1 \leq U$.
- (iii) $U^\perp = \ker(\varphi_R)$ and ${}^\perp V = \ker(\varphi_L)$.

Definition 6.8 – Degenerate bilinear form

φ is degenerate if $U^\perp$ or ${}^\perp V$ is nontrivial.

Proposition 6.3

Let U, V be finite-dimensional and B, C bases for U, V respectively. φ is non-degenerate if and only if $\dim(U) = \dim(V)$ and $[\varphi]_{B,C}$ is invertible.

Proof. Non-degeneracy is equivalent to $\ker(\varphi_L)$ and $\ker(\varphi_R)$ both trivial. This is true if and only if $\dim(U) = \text{rk}(\varphi_L) = \text{rk}(\varphi) = \text{rk}(\varphi_R) = \dim(V)$ by Corollary 6.2. This is equivalent to $[\varphi]_{B,C}$ being a square matrix of full rank. □

Definition 6.9 – Symmetric bilinear form

A bilinear form $\varphi : V \times V \rightarrow \mathbb{F}$ is symmetric if for all $u, v \in V$,

$$\varphi(u, v) = \varphi(v, u).$$

- (i) $A \in \mathcal{M}_{n \times n}(\mathbb{F})$ is symmetric if $A^\top = A$. For finite-dimensional V , a bilinear form φ is symmetric if and only if for some (or equivalently, any) basis B for V , $[\varphi]_{B,B}$ is symmetric.
- (ii) If a bilinear form φ is symmetric, then for any $S \subseteq V$, ${}^\perp S = S^\perp$. In particular, the left kernel is equal to the right kernel, and we call just call it the kernel of φ .

Definition 6.10 – Quadratic form

A quadratic form on V is a function $Q : V \rightarrow \mathbb{F}$ such that there exists a bilinear form $\varphi : V \times V \rightarrow \mathbb{F}$ such that for all $v \in V$,

$$Q(v) = \varphi(v, v).$$

Example 6.2

Take $V = \mathbb{R}^2$. For $A \in \mathcal{M}_{2 \times 2}(\mathbb{R})$, define

$$Q_A(\mathbf{x}) = \mathbf{x}^\top A \mathbf{x} = a_{11}x^2 + (a_{12} + a_{21})xy + a_{22}y^2.$$

The corresponding bilinear form is $\varphi(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top A \mathbf{y}$. Note that $Q_A = Q_{\frac{1}{2}(A+A^\top)}$.

Proposition 6.4 – Polarisation identity

Suppose $1 + 1 \neq 0$ in $\mathbb{F}$. Let $Q : V \rightarrow \mathbb{F}$ be a quadratic form. There is a unique symmetric bilinear form $\psi : V \times V \rightarrow \mathbb{F}$ such that $Q(v) = \psi(v, v)$ for all $v \in V$. ψ is given by

$$\psi(u, v) = \frac{1}{2}(Q(u+v) - Q(u) - Q(v)).$$

Proof. For existence, let φ be as in the definition of the quadratic form. Set $\psi(u, v) = (\varphi(u, v) + \varphi(v, u))/2$. For uniqueness, suppose ψ is a symmetric bilinear form satisfying $Q(v) = \psi(v, v)$ for all $v \in V$. Then,

$$\begin{aligned} Q(u+v) &= \psi(u+v, u+v) \\ &= \psi(u, u) + \psi(u, v) + \psi(v, u) + \psi(v, v) \\ &= Q(u) + Q(v) + 2\psi(u, v) \end{aligned}$$

which gives the result. □

Theorem 6.1

Suppose $1 + 1 \neq 0$ in $\mathbb{F}$ and V is a finite-dimensional $\mathbb{F}$ -vector space. Let $\varphi : V \times V \rightarrow \mathbb{F}$ be a symmetric bilinear form. There is a basis B for V such that $[\varphi]_{B,B}$ is diagonal.

Proof. We proceed by induction on $n = \dim(V)$. If for all $v \in V$, $\varphi(v, v) = 0$, then by polarisation, $\varphi = 0$ and we are done. If not, there exists $0 \neq v_1 \in V$ such that $\varphi(v_1, v_1) \neq 0$. Set $U = \langle v_1 \rangle^\perp = \ker(\varphi_L(v_1) : V \rightarrow \mathbb{F})$. Then $v_1 \notin U$, so by rank-nullity, $\dim(U) = n - 1$ and $V = \langle v_1 \rangle \oplus U$. $\varphi|_{U \times U} : U \times U \rightarrow \mathbb{F}$ is a symmetric bilinear form so by induction, there exists a basis $B_U = \{v_2, \dots, v_n\}$ for U such that $[\varphi|_{U \times U}]_{B_U, B_U}$ is diagonal. Then, $B = \{v_1, \dots, v_n\}$ is a basis for V and

$$[\varphi]_{B,B} = \begin{pmatrix} \varphi(v_1, v_1) & & 0 \\ & \ddots & \\ 0 & & \varphi(v_n, v_n) \end{pmatrix}.$$

□

The assumption that φ is symmetric is necessary. If $P^\top AP = D$ is diagonal with $P \in \text{GL}_n(\mathbb{F})$, then $P^\top AP = D^\top = (P^\top AP)^\top = P^\top A^\top P$ so $A = A^\top$. Proposition 6.4 yields an algorithm for finding a diagonal basis.

Corollary 6.3

Suppose $1 + 1 \neq 0$ in $\mathbb{F}$. Every symmetric matrix is congruent to a diagonal matrix.

Example 6.3

In practice, shortcuts are available. Take $\mathbf{x} \in \mathbb{R}^3$ and

$$\begin{aligned} Q(\mathbf{x}) &= x_1^2 - 4x_2x_3 + x_3^2 \\ &= x_1^2 + (x_3 - 2x_2)^2 - (2x_2)^2 \\ &= z_1^2 + z_2^2 - z_3^2 \end{aligned}$$

where

$$\mathbf{z} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & -2 & 1 \\ 0 & 2 & 0 \end{pmatrix}}_{= P^{-1}} \mathbf{x}.$$

$Q(\mathbf{x}) = \psi(\mathbf{x}, \mathbf{x})$ where $\psi(\mathbf{x}, \mathbf{y}) = x_1y_1 - 2(x_2y_3 + x_3y_2) + x_3y_3$. With respect to the standard basis, ψ has matrix

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -2 \\ 0 & -2 & 1 \end{pmatrix} \text{ and } P^\top AP = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Corollary 6.4

Suppose V, φ are as in Theorem 6.1 and $Q(v) = \varphi(v, v)$. There is a basis $B = \{v_1, \dots, v_n\}$ for V such that

(i) $[\varphi]_{B,B} = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$ with $r = \text{rk}(\varphi)$, or equivalently, $Q\left(\sum_{i=1}^n \lambda_i v_i\right) = \sum_{i=1}^r \lambda_i^2$ if $\mathbb{F} = \mathbb{C}$.

(ii) $[\varphi]_{B,B} = \begin{bmatrix} I_p & 0 & 0 \\ 0 & -I_q & 0 \\ 0 & 0 & 0 \end{bmatrix}$ with $p + q = \text{rk}(\varphi)$, or equivalently, $Q\left(\sum_{i=1}^n \lambda_i v_i\right) = \sum_{i=1}^p \lambda_i^2 - \sum_{i=p+1}^{p+q} \lambda_i^2$ if $\mathbb{F} = \mathbb{R}$.

Proof. Pick a basis $C = \{c_1, \dots, c_n\}$ such that

$$[\varphi]_{C,C} = \begin{pmatrix} a_1 & & 0 \\ & \ddots & \\ 0 & & a_n \end{pmatrix}.$$

We can reorder the indices such that $a_i \neq 0$ if and only if $i \leq r = \text{rk}(\varphi)$. If $\mathbb{F} = \mathbb{R}$, we can make $a_i > 0$ for $i \leq p$ and $a_i < 0$ for $p + 1 \leq i \leq p + q$ with $r = p + q$. If $\mathbb{F} = \mathbb{C}$, set

$$v_i = \begin{cases} c_i / \sqrt{a_i} & \text{if } i \leq r \\ c_i & \text{if } i \geq r + 1. \end{cases}$$

If $\mathbb{F} = \mathbb{R}$, set

$$v_i = \begin{cases} c_i/\sqrt{a_i} & \text{if } i \geq p \\ c_i/\sqrt{-a_i} & \text{if } p+1 \leq i \leq p+q \\ c_i & \text{if } i \geq p+q+1. \end{cases}$$

□

Definition 6.11 – Positive/Negative (semi-)definite

Let $\varphi : V \times V \rightarrow \mathbb{R}$ be a symmetric bilinear form.

- (i) φ is positive definite if $\varphi(v, v) > 0$ for all $\underline{0} \neq v \in V$.
- (ii) φ is positive semi-definite if $\varphi(v, v) \geq 0$ for all $\underline{0} \neq v \in V$.
- (iii) φ is negative definite if $\varphi(v, v) < 0$ for all $\underline{0} \neq v \in V$.
- (iv) φ is negative semi-definite if $\varphi(v, v) \leq 0$ for all $\underline{0} \neq v \in V$.

If none of these are satisfied, we say that φ is indefinite. Similar terminology applies to quadratic forms.

Example 6.4

Let V be finite-dimensional and B be a basis for V such that

$$[\varphi]_{B,B} = \begin{bmatrix} I_p & 0 \\ 0 & 0 \end{bmatrix}.$$

Then, φ is positive semi-definite, and it is positive definite if and only if $p = \dim(V)$.

Theorem 6.2 – Sylvester's law of inertia

If V is finite-dimensional and φ is a symmetric bilinear form on V . If B, B' are bases for V such that

$$[\varphi]_{B,B} = \begin{bmatrix} I_p & 0 & 0 \\ 0 & -I_q & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ and } [\varphi]_{B',B'} = \begin{bmatrix} I_{p'} & 0 & 0 \\ 0 & -I_{q'} & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

then $p = p'$ and $q = q'$.

Proof. We need the following lemma.

Lemma 6.2

Let $U, W \leq V$.

- (i) If $\varphi|_U$ is positive definite and $\varphi|_W$ is negative semi-definite, then $U \cap W = \{\underline{0}\}$.
- (ii) Similarly, if $\varphi|_U$ is negative definite and $\varphi|_W$ is positive semi-definite, then $U \cap W = \{\underline{0}\}$.

Proof. For all $\underline{0} \neq v \in U \cap W$, we have $\varphi(v, v) > 0$ since $v \in U$ and $\varphi(v, v) \leq 0$ since $v \in W$. Hence, there is no such v . Similar for (ii). □

We will show that p is the maximal dimension of a subspace on which φ is positive definite, and q is the

maximal dimension of a subspace on which φ is negative definite. Let $B = \{v_1, \dots, v_n\}$ be such that

$$[\varphi]_{B,B} = \begin{bmatrix} I_p & 0 & 0 \\ 0 & -I_q & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Set $U = \langle v_1, \dots, v_p \rangle$ and $W = \langle v_{p+1}, \dots, v_n \rangle$ so $\varphi|_U$ is positive definite and $\varphi|_W$ is negative semi-definite. If there exists $U' \leq V$ such that $\varphi|_{U'}$ is positive definite and $\dim(U') > p$, then by the sum-intersection formula, $\dim(U' \cap W) \geq 1$, which contradicts with Lemma 6.2. The case for q is similar. $\square$

Definition 6.12 – Signature

The signature of the φ is $\sigma(\varphi) = p - q$.

Corollary 6.5

$\sigma(\varphi)$ is well-defined.

Definition 6.13 – Totally isotropic

$T \leq V$ is totally isotropic for φ if for all $t, u \in T$, we have $\varphi(t, u) = 0$.

If $T, U \leq V$ with T totally isotropic and U positive/negative definite, then $U \cap T = \{0\}$.

Proposition 6.5

If V is finite-dimensional, the maximal dimensional of a totally isotropic subspace of V for a symmetric bilinear form Q is $\dim(V) - \max\{p, q\}$ where p, q are as in Theorem 6.2.

Proof. Let $\dim(V) = n$ and $B = \{v_1, \dots, v_n\}$ be a basis such that

$$[\varphi]_{B,B} = \begin{bmatrix} I_p & 0 & 0 \\ 0 & -I_q & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Without loss of generality, assume $p \geq q$. Let $T = \langle v_1 + v_{p+1}, v_2 + v_{p+2}, \dots, v_q + v_{p+q}, v_{p+q+1}, \dots, v_n \rangle$. Then, T is a totally isotropic subspace with $\dim(T) = n - p$. Suppose T' is another totally isotropic subspace with $\dim(T') > n - p$. Set $U = \langle v_1, \dots, v_p \rangle$, then $\varphi|_U$ is positive definite. By the sum-intersection formula, $\dim(U \cap T') = 1$ so $U \cap T' \neq \{0\}$ which is a contradiction. $\square$

Definition 6.14 – Sesquilinear form

Let V and W be $\mathbb{C}$ -vector spaces. A sesquilinear form is a function $\varphi : V \times W \rightarrow \mathbb{C}$ such that

$$(i) \quad \varphi(\lambda v_1 + v_2, w_1) = \lambda \varphi(v_1, w_1) + \varphi(v_2, w_1)$$

$$(ii) \quad \varphi(v_1, \mu w_1 + w_2) = \bar{\mu} \varphi(v_1, w_1) + \varphi(v_1, w_2)$$

for all $\lambda, \mu \in \mathbb{C}$ and $v_1, v_2 \in V$ and $w_1, w_2 \in W$.

Example 6.5

The standard dot product on $\mathbb{C}^n$, which is $\mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^n x_i \bar{y}_i$ is not bilinear but is instead sesquilinear.

Definition 6.15

Let V, W be finite-dimensional and $\varphi : V \times W \rightarrow \mathbb{C}$ be a sesquilinear form. Let $B = \{b_1, \dots, b_m\}$ and $C = \{c_1, \dots, c_n\}$ be bases for V and W respectively.

$$[\varphi]_{B,C} = (\varphi(b_i, c_j))_{ij} \in \mathcal{M}_{m \times n}(\mathbb{C})$$

is the matrix of φ with respect to B and C .

Proposition 6.6

$[\varphi]_{B,C}$ is the unique matrix satisfying

$$\varphi(v, w) = ([v]_B)^\top [\varphi]_{B,C} \overline{[w]_C}$$

for all $v \in V$ and $w \in W$.

Corollary 6.6

Let B, B' be bases for V and C, C' be bases for W . Then,

$$[\varphi]_{B',C'} = ([\text{id}_V]_B^{B'})^\top [\varphi]_{B,C} \overline{[\text{id}_W]_C^{C'}}.$$

Proof. The same as for bilinear forms. □

Definition 6.16 – Hermitian form

A sesquilinear form $\varphi : V \times V \rightarrow \mathbb{C}$ is Hermitian if for all $u, v \in V$, we have $\varphi(u, v) = \overline{\varphi(v, u)}$.

Note that for all $v \in V$, $\varphi(v, v) \in \mathbb{R}$. Also, for $\lambda \in \mathbb{C}$, $\varphi(\lambda v, \lambda v) = |\lambda|^2 \varphi(v, v)$. Thus, we can define positive/negative (semi-)definite Hermitian forms. Moreover, $\varphi(u, v) = 0$ if and only if $\varphi(v, u) = 0$. Thus, for any $S \subseteq V$, ${}^\perp S = S^\perp$.

Definition 6.17 – Hermitian matrix

$A \in \mathcal{M}_{n \times n}(\mathbb{C})$ is Hermitian if $A = \overline{A}^\top$.

Lemma 6.3

Let V be finite-dimensional. A sesquilinear form $\varphi : V \times V \rightarrow \mathbb{C}$ is Hermitian if and only if for some (or equivalently, any) basis B for V , $[\varphi]_{B,B}$ is Hermitian.

Proof. If φ is Hermitian, let $B = \{b_1, \dots, b_n\}$ be a basis for V and $A = [\varphi]_{B,B}$.

$$a_{ij} = \varphi(b_i, b_j) = \overline{\varphi(b_j, b_i)} = \overline{a_{ji}}$$

so A is Hermitian.

If $[\varphi]_{B,B}$ is Hermitian, for any $v, w \in V$,

$$\varphi(v, w) = [v]_B^\top A \overline{[w]_B} = \overline{[w]_B^\top A^\top [v]_B} = \overline{[w]_B^\top \overline{A}^\top [v]_B} = \overline{\varphi(w, v)}$$

so φ is sesquilinear. □

Proposition 6.7

Let V be a $\mathbb{C}$ -vector space. A Hermitian form φ on V is uniquely determined by $Q : V \rightarrow \mathbb{R}$ given by $Q(v) = \varphi(v, v)$ via

$$\varphi(u, v) = \frac{1}{4}(Q(u+v) - Q(u-v) + iQ(u+iv) - iQ(u-iv)).$$

Proof. On example sheet 4. □

Theorem 6.3

Let V be a finite-dimensional $\mathbb{C}$ -vector space and φ be a Hermitian form on V .

(i) There is a basis B for V such that

$$[\varphi]_{B,B} = \begin{bmatrix} I_p & 0 & 0 \\ 0 & -I_q & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

(ii) p and q are uniquely determined by φ .

Proof. Only a sketch proof here. It is almost identical to the real symmetric case. For existence, we proceed by induction on $\dim(V)$. If $\varphi = 0$, then we are done. If not, by the polarisation identity, there exists $0 \neq v_1 \in V$ such that $\varphi(v_1, v_1) \neq 0$. Let $U = \langle v_1 \rangle^\perp$. We can check that $V = \langle v_1 \rangle \oplus U$ and apply induction on $\varphi|_U$. □

Definition 6.18 – Inner product space

Let V be a $\mathbb{R}$ -vector space (or respectively, a $\mathbb{C}$ -vector space). An inner product on V is a positive definite symmetric bilinear form (respectively, a Hermitian form) φ on V . The pair (V, φ) is an inner product space. We write $\langle v, w \rangle$ for $\varphi(v, w)$.

Example 6.6

(i) The dot product on $\mathbb{R}^n$ (or on $\mathbb{C}^n$) is an inner product.

(ii) Let $V = C([0, 1])$ be a $\mathbb{R}$ -vector space. $\langle f, g \rangle = \int_0^1 f(t)g(t) dt$ is an inner product. More generally, given a continuous weight function $w : [0, 1] \rightarrow (0, \infty)$,

$$\langle f, g \rangle_w = \int_0^1 f(t)g(t)w(t) dt$$

is an inner product. If V is a $\mathbb{C}$ -vector space, we simply conjugate $g(t)$ in the integral.

Definition 6.19 – Length and distance

Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space. The length of $v \in V$ is

$$\|v\| = \sqrt{\langle v, v \rangle}.$$

The distance between $v \in V$ and $w \in V$ is $\|v - w\|$.

0 is the only vector of length zero. Also note that $\|\cdot\| : V \rightarrow \mathbb{R}_{\geq 0}$ determines $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{F}$ uniquely by the polarisation identity.

Lemma 6.4 – Cauchy-Schwarz inequality

For all $v, w \in V$, we have

$$|\langle v, w \rangle| \leq \|v\| \|w\|.$$

Proof. If $v = \underline{0}$, then we are done. If not, for $t \in \mathbb{F}$,

$$\begin{aligned} 0 &\leq \|tv + w\|^2 \\ &= \langle tv + w, tv + w \rangle \\ &= |t|^2 \langle v, v \rangle + t \langle v, w \rangle + \bar{t} \langle w, v \rangle + \langle w, w \rangle \\ &= |t|^2 \|v\|^2 + t \langle v, w \rangle + \bar{t} \overline{\langle v, w \rangle} + \|w\|^2. \end{aligned}$$

Set $t = -\overline{\langle v, w \rangle} / \|v\|^2$, so

$$0 \leq -\frac{|\langle v, w \rangle|^2}{\|v\|^2} + \|w\|^2$$

which gives the result. $\square$

From the proof, we have equality if and only if $\dim(\langle v, w \rangle) \leq 1$. Moreover, for $\mathbb{F} = \mathbb{R}$, we can define the angle θ from v to w by

$$\cos \theta = \frac{\langle v, w \rangle}{\|v\| \|w\|} \in [-1, 1].$$

Corollary 6.7 – Triangle inequality

For all $v, w \in V$, we have

$$\|v + w\| \leq \|v\| + \|w\|.$$

Proof.

$$\begin{aligned} \|v + w\|^2 &= \langle v + w, v + w \rangle \\ &= \|v\|^2 + \|w\|^2 + \langle v, w \rangle + \overline{\langle v, w \rangle} \\ &\leq \|v\|^2 + \|w\|^2 + 2\|v\| \|w\| \\ &= (\|v\| + \|w\|)^2 \end{aligned}$$

$\square$

Definition 6.20 – Orthogonal and orthonormal sets

A subset $E \subseteq V$ is orthogonal if for all $d, e \in E$ with $d \neq e$, we have $\langle d, e \rangle = 0$. E is orthonormal if it is orthogonal and all $e \in E$ satisfy $\|e\| = 1$.

If V is finite-dimensional and B is a basis for V with $[\langle \cdot, \cdot \rangle]_B = I_n$, then B is an orthonormal basis for V .

Lemma 6.5

If $E \subseteq V$ is orthogonal and $\underline{0} \notin E$, then

- (i) E is linearly independent
- (ii) for any $v \in \langle E \rangle$, we have

$$v = \sum_{e \in E} \frac{\langle v, e \rangle}{\langle e, e \rangle} e.$$

Proof. For $v \in \langle E \rangle$, write $v \in \sum'_{e \in E} \lambda_e e$. Then, for $d \in E$,

$$\langle v, d \rangle = \sum'_{e \in E} \langle e, d \rangle \lambda_e = \langle d, d \rangle \lambda_d$$

so $\lambda_d = \langle v, d \rangle / \langle d, d \rangle$. In particular, if $v = \underline{0}$, then $\lambda_d = 0$. □

Corollary 6.8 – Parseval's identity

If E is an orthonormal basis for V , then for all $v, w \in V$,

$$\langle v, w \rangle = \sum'_{e \in E} \langle v, e \rangle \overline{\langle w, e \rangle}.$$

Proof.

$$\langle v, w \rangle = \left\langle \sum'_{e \in E} \langle v, e \rangle e, \sum'_{d \in E} \langle w, d \rangle d \right\rangle = \sum'_{d, e \in E} \langle v, e \rangle \overline{\langle w, d \rangle} \langle e, d \rangle = \sum'_{e \in E} \langle v, e \rangle \overline{\langle w, e \rangle}$$

□

Theorem 6.4 – Gram-Schmidt process

Let V be an inner product space. Let $\{v_1, \dots, v_n\} \subseteq V$ be linearly independent. There is an orthonormal set $\{e_1, \dots, e_n\} \subseteq V$ such that $\langle e_1, \dots, e_k \rangle = \langle v_1, \dots, v_k \rangle$ for all k .

Proof. We proceed by induction on k . If $k = 1$ and $v_1 \neq \underline{0}$, set $e_1 = v_1 / \|v_1\|$. Suppose we have found $e_1, \dots, e_k$. Set $\tilde{e}_{k+1} = v_{k+1} - \sum_{i=1}^k \langle v_{k+1}, e_i \rangle e_i$. Then, $\tilde{e}_{k+1} \neq \underline{0}$ as $v_{k+1} \notin \langle v_1, \dots, v_k \rangle = \langle e_1, \dots, e_k \rangle$. Also, by taking the inner product of $\tilde{e}_{k+1}$ with some e_j , we see that $\tilde{e}_{k+1}$ is orthogonal to e_j for all $j \leq k$. Moreover, $\langle e_1, \dots, e_k, \tilde{e}_{k+1} \rangle = \langle v_1, \dots, v_{k+1} \rangle$. Set $e_{k+1} = \tilde{e}_{k+1} / \|\tilde{e}_{k+1}\|$ and we are done. □

Note that if $\{v_1, \dots, v_k\}$ are already orthonormal, then $v_i = e_i$ for all $i \leq k$.

Corollary 6.9

If V is a finite-dimensional inner product space, for any orthonormal set E in V , there is an orthonormal basis containing E .

Proof. As E is orthonormal, it is linearly independent. We can extend it to a basis B . Apply the Gram-Schmidt process to B , and E is unchanged by the process. □

Definition 6.21 – Orthogonal direct sum

Let V be an inner product space and $U, W \leq V$. V is the orthogonal direct sum of U and W if

- (i) $U + W = V$
- (ii) for all $u \in U$ and $w \in W$, we have $\langle u, w \rangle = 0$.

We write $V = U \oplus W$.

(ii) implies $U \cap W = \{\underline{0}\}$. Hence, $V = U \oplus W$ implies $V = U \oplus W$.

Lemma 6.6

If V is finite-dimensional and $W \leq V$, then $V = W \oplus W^\perp$.

Proof. Since W and $W^\perp$ satisfies (ii), it suffices to check that $V = W + W^\perp$. Let E_W be an orthonormal basis for W . Extend to an orthonormal basis E_V for V . Then,

$$V = \langle E_W \cup (E_V \setminus E_W) \rangle = W + \langle E_V \setminus E_W \rangle.$$

Also, $E_V \setminus E_W \subseteq W^\perp$. □

Corollary 6.10

$$\dim(W^\perp) = \dim(V) - \dim(W)$$

Proof. $V = W \oplus W^\perp$ □

Definition 6.22 – Orthogonal projection

Let V be a finite-dimensional inner product space and $W \leq V$. The orthogonal projection $\pi_W : V \rightarrow V$ of V onto W is given by $\pi_W(v) = w$, where $v = w + u$ with $w \in W$ and $u \in W^\perp$ both unique, since $V = W \oplus W^\perp$.

- (i) π_W is linear. $\text{im}(\pi_W) = W$ and $\ker(\pi_W) = W^\perp$.
- (ii) $\pi_W|_W = \text{id}_W$ and $\pi_W^2 = \pi_W$.
- (iii) $\text{id}_V - \pi_W = \pi_{W^\perp}$.

Proposition 6.8 – Closest-point projection

For all $v \in V$ and $w \in W$,

$$\|v - \pi_W(v)\| \leq \|v - w\|$$

with equality if and only if $w = \pi_W(v)$.

Proof.

$$\|v - w\|^2 = \|\underbrace{(v - \pi_W(v))}_{\in W^\perp} + \underbrace{(\pi_W(v) - w)}_{\in W}\|^2 = \|v - \pi_W(v)\|^2 + \|\pi_W(v) - w\|^2 \geq \|v - \pi_W(v)\|^2$$

with equality if and only if $\|\pi_W(v) - w\| = 0$. □

Definition 6.23 – External orthogonal direct sum

Let $(V_1, \langle \cdot, \cdot \rangle_1)$ and $(V_2, \langle \cdot, \cdot \rangle_2)$ be inner product spaces. Define $\langle \cdot, \cdot \rangle$ on $V_1 \times V_2$ via

$$\langle (v_1, v_2), (v'_1, v'_2) \rangle = \langle v_1, v'_1 \rangle_1 + \langle v_2, v'_2 \rangle_2.$$

Then, $\langle \cdot, \cdot \rangle$ is an inner product on $V_1 \oplus V_2$.

Let $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. Let $\mathbf{v} \in \mathbb{F}^n$, $\mathbf{w} \in \mathbb{F}^m$ and $A \in \mathcal{M}_{m \times n}(\mathbb{F})$. Then,

$$(A\mathbf{v}) \cdot \mathbf{w} = (A\mathbf{v})^\top \overline{\mathbf{w}} = (\mathbf{v}^\top A^\top) \overline{\mathbf{w}} = \mathbf{v}^\top \overline{(A^\top \mathbf{w})} = \mathbf{v} \cdot (\overline{A^\top \mathbf{w}}).$$

Theorem 6.5

Let V and W be finite-dimensional inner product spaces. Let $\alpha \in \mathcal{L}(V, W)$.

(i) There exists a unique linear map $\alpha^* : W \rightarrow V$ satisfying

$$\langle \alpha(v), w \rangle_W = \langle v, \alpha^*(w) \rangle_V$$

for all $v \in V$ and $w \in W$.

(ii) For any orthonormal bases B and C for V and W respectively, we have $[\alpha^*]_B^C = \overline{[\alpha]_C^B}^\top$.

Definition 6.24 – Adjoint map

α^* as defined above is the adjoint map of α .

Proof. Let $B = \{b_1, \dots, b_n\}$ and $C = \{c_1, \dots, c_m\}$. Let $A = [\alpha]_C^B$. Define the linear map $\alpha^* : W \rightarrow V$ by $\alpha^*(c_j) = \sum_{i=1}^n \overline{a_{ji}} b_i$ so that $[\alpha^*]_B^C = \overline{A}^\top$. Then,

$$\begin{aligned} \langle \alpha(v), w \rangle_W &= [\alpha(v)]_C^\top [\overline{w}]_C \\ &= (A[v]_B)^\top [\overline{w}]_C \\ &= [v]_B^\top A^\top [\overline{w}]_C \\ &= [v]_B^\top \overline{A^\top [w]_C} \\ &= [v]_B^\top \overline{([\alpha^*]_B^C [w]_C)} \\ &= [v]_B^\top \overline{[\alpha^*(w)]_B} = \langle v, \alpha^*(w) \rangle_V \end{aligned}$$

since $[\langle \cdot, \cdot \rangle_V]_B = I_n$ and $[\langle \cdot, \cdot \rangle_W]_C = I_m$. For uniqueness, suppose $\alpha' : W \rightarrow V$ satisfies the property. Let $A' = [\alpha']_B^C$. Then, as above, $\langle \alpha(b_i), c_j \rangle_W = \langle b_i, \alpha'(c_j) \rangle_V$. However,

$$\langle \alpha'(b_i), c_j \rangle_W = (A\mathbf{e}_i)^\top \overline{\mathbf{e}_j} = a_{ji} \text{ and } \langle b_i, \alpha'(c_j) \rangle_V = \mathbf{e}_i^\top \overline{(A'\mathbf{e}_j)} = \overline{a'_{ij}}.$$

Hence, $[\alpha']_B^C = A' = \overline{A}^\top = [\alpha^*]_B^C$ so $\alpha' = \alpha^*$. $\square$

Let V be any $\mathbb{R}$ -inner product space. Recall that for $v \in V$, we have $\varphi_R(v) \in V^*$ given by $(\varphi_R(v))(u) = \langle u, v \rangle$. Moreover, $\varphi_R : V \rightarrow V^*$ is linear.

Theorem 6.6 – Riesz representation theorem

For V a finite-dimensional $\mathbb{R}$ -inner product space, $\varphi_R : V \rightarrow V^*$ is a linear isomorphism.

Proof. For linearity, see Lemma 6.1. This makes use of the fact that V is real. For injectivity, suppose that $v \in \ker(\varphi_R)$, then $\varphi_R(v) = \mathbf{0}$ in V^* . So, $0 = (\varphi_R(v))(v) = \langle v, v \rangle$. Hence, $v = 0$. Surjectivity is immediate from the rank-nullity theorem, as $\dim(V) = \dim(V^*)$. $\square$

We could also define the adjoint map as the composition $W \xrightarrow{\varphi_{R,W}} W^* \xrightarrow{\alpha^* \text{ (dual)}} V^* \xrightarrow{\varphi_{R,V}^{-1}} V$.

Definition 6.25 – Self-adjoint

$\alpha \in \mathcal{L}(V, V)$ is self-adjoint if $\alpha^* = \alpha$.

Lemma 6.7

If $\mathbb{F} = \mathbb{R}$ (or respectively, $\mathbb{C}$), α is self-adjoint if and only if for some (or equivalently, any) orthonormal basis B for V , $[\alpha]_B^B$ is symmetric (respectively, Hermitian).

Proof. Fix B an orthonormal basis. α is self-adjoint $\iff [\alpha^*]_B^B = [\alpha]_B^B \iff \overline{([\alpha]_B^B)}^\top = [\alpha]_B^B$. $\square$

Definition 6.26 – Isometry

$\alpha \in \mathcal{L}(V, W)$ is an isometry if for all $u, v \in U$,

$$\langle \alpha(u), \alpha(v) \rangle_W = \langle u, v \rangle_V.$$

- (i) By polarisation, $\alpha \in \mathcal{L}(V, W)$ is an isometry if and only if for all $v \in V$, $\|\alpha(v)\|_W = \|v\|_V$.
- (ii) Every isometry is injective.
- (iii) If α is an isometry and $B \subseteq V$ is orthonormal, then $\alpha(B) \subseteq W$ is also orthonormal.

Proposition 6.9

Suppose $\alpha \in \mathcal{L}(V, W)$ is invertible. α is an isometry if and only if $\alpha^{-1} = \alpha^*$.

Proof. Note that $\langle \alpha(u), \alpha(v) \rangle_W = \langle u, \alpha^*(\alpha(v)) \rangle_V$. If $\alpha^* = \alpha^{-1}$, then $\langle \alpha(u), \alpha(v) \rangle = \langle u, v \rangle$, so it is an isometry. If α is an isometry, that is, $\langle \alpha(u), \alpha(v) \rangle = \langle u, v \rangle$ for all $u, v \in V$, then $\langle u, \alpha^*(\alpha(v)) - v \rangle = 0$ for all $u, v \in V$. Set $u = \alpha^*(\alpha(v)) - v$, then $\alpha^*(\alpha(v)) = v$ for all $v \in V$. Set $v = \alpha^{-1}(w)$ with $w \in W$, then $\alpha^*(w) = \alpha^{-1}(w)$ for all $w \in W$. $\square$

Corollary 6.11

Let $\mathbb{F} = \mathbb{R}$ (or respectively, $\mathbb{C}$) and V be finite-dimensional. $\alpha \in \mathcal{L}(V, V)$ is an isometry if and only if for some (or equivalently, any) orthonormal basis B for V , $[\alpha]_B^B$ is an orthogonal (respectively, unitary) matrix.

Proof. α is an isometry $\iff \alpha \circ \alpha^{-1} = \text{id}_V \iff [\alpha]_B^B [\alpha^*]_B^B = I_n \iff [\alpha]_B^B \overline{[\alpha]_B^B}^\top = I_n$. $\square$

From IA Groups, we defined $O(n) = \{A \in \mathcal{M}_{n \times n}(\mathbb{R}) : AA^\top = I_n\}$ as the orthogonal group and $U(n) = \{A \in \mathcal{M}_{n \times n}(\mathbb{C}) : A\bar{A}^\top = I_n\}$ as the unitary group.

Definition 6.27 – Orthogonal and unitary group

For $\mathbb{F} = \mathbb{R}$, define $O(V) = \{\alpha \in \mathcal{L}(V, V) : \alpha \text{ is an isometry}\}$ to be the orthogonal group of V .
For $\mathbb{F} = \mathbb{C}$, define $U(V) = \{\alpha \in \mathcal{L}(V, V) : \alpha \text{ is an isometry}\}$ to be the unitary group of V .

Let $\mathbb{F} = \mathbb{R}$ (or respectively, $\mathbb{C}$). Let $n = \dim(V)$ and B be an orthonormal basis for V . $\alpha \mapsto [\alpha]_B^B$ defines an isomorphism of groups $O(V) \rightarrow O(n)$ (respectively, $U(V) \rightarrow U(n)$). Note that for $A \in \mathcal{M}_{n \times n}(\mathbb{F})$, $A \in O(n)$ or $U(n)$ if and only if the columns of A form an orthonormal basis for $\mathbb{F}^n$.

Let $T_n(\mathbb{F}) = \{A \in GL_n(\mathbb{F}) : A \text{ is upper triangular}\} \leq GL_n(\mathbb{F})$. Then, if $\mathbb{F} = \mathbb{R}$, we have $GL_n(\mathbb{R}) = O(n) \cdot T_n(\mathbb{R})$. If $\mathbb{F} = \mathbb{C}$, we have $GL_n(\mathbb{C}) = U(n) \cdot T_n(\mathbb{C})$. To see this, apply the Gram-Schmidt process to the columns of A .

Example 6.7

For $W \leq V$, recall that $V = W \oplus W^\perp$ and we can define π_W . Then,

$$\langle w + u, \pi_W(w' + u') \rangle = \langle w + u, w' \rangle = \langle w, w' \rangle = \langle w, w' + u' \rangle = \langle \pi_W(w + u), w' + u' \rangle$$

so $\pi_W^* = \pi_W$. Recall also that $\pi_W^2 = \pi_W$. Conversely, suppose $\alpha \in \mathcal{L}(V, V)$ is self-adjoint with $\alpha^2 = \alpha$. By example sheet 1 question 4, $V = \text{im}(\alpha) \oplus \ker(\alpha)$. Moreover, for $w = \alpha(v) \in \text{im}(\alpha)$ and $u \in \ker(\alpha)$,

$$\langle w, u \rangle = \langle \alpha(v), u \rangle = \langle v, \alpha(u) \rangle = \langle v, 0 \rangle = 0$$

so $V = \text{im}(\alpha) \oplus \ker(\alpha)$. Finally, $\alpha(\alpha(v) + u) = \alpha^2(v) = \alpha(v)$ so α is an orthogonal projection onto $\text{im}(\alpha)$.

Theorem 6.7 – Spectral theorem for self-adjoint endomorphisms

Let V be a finite-dimensional inner product space and $\alpha \in \mathcal{L}(V, V)$. If α is self-adjoint, then V has an orthonormal basis consisting of eigenvectors of α with real eigenvalues.

Proof. For the proof, we need three lemmas.

Lemma 6.8

Let α be self-adjoint.

- (i) All eigenvalues of α are real.
- (ii) Eigenvectors of distinct eigenvalues are orthogonal.

Proof.

- (i) Suppose $0 \neq v \in V_\lambda$, then

$$\lambda \langle v, v \rangle = \langle \lambda v, v \rangle = \langle \alpha(v), v \rangle = \langle v, \alpha(v) \rangle = \langle v, \lambda v \rangle = \bar{\lambda} \langle v, v \rangle.$$

Since $\langle v, v \rangle \neq 0$, we must have $\lambda = \bar{\lambda}$.

- (ii) Suppose $\lambda \neq \mu$, $\alpha(v) = \lambda v$ and $\alpha(w) = \mu w$. Then,

$$\lambda \langle v, w \rangle = \langle \alpha(v), w \rangle = \langle v, \alpha(w) \rangle = \langle v, \mu w \rangle = \mu \langle v, w \rangle$$

as $\mu \in \mathbb{R}$. Since $\lambda \neq \mu$, we must have $\langle v, w \rangle = 0$.

□

Corollary 6.12

If $A \in \mathcal{M}_{n \times n}(\mathbb{C})$ is Hermitian, then every eigenvalue of A is real.

Lemma 6.9

For any $\alpha \in \mathcal{L}(V, V)$ and $W \leq V$, if $\alpha(W) \leq W$, then $\alpha^*(W^\perp) \leq W^\perp$.

Proof. Let $w \in W$ and $u \in W^\perp$. Then, $\langle w, \alpha^*(u) \rangle = \langle \alpha(w), u \rangle = 0$ as $\alpha(w) \in W$. Since this holds for all $w \in W$, we must have $\alpha^*(u) \in W^\perp$ for all $u \in W^\perp$. $\square$

Lemma 6.10

If α is self-adjoint, then α has an eigenvector.

Proof. Let B be an orthonormal basis for V and $A = [\alpha]_B^B$. Consider $\chi_A(t)$ as a complex polynomial. By the fundamental theorem of algebra, χ_A has a root $\lambda \in \mathbb{C}$. If $\mathbb{F} = \mathbb{C}$, then we are done as λ is an eigenvalue of α . If $\mathbb{F} = \mathbb{R}$, then A is symmetric and so is Hermitian, and λ is an eigenvalue of α . By Corollary 6.12, $\lambda \in \mathbb{R}$. Let $\underline{0} \neq c \in \mathbb{C}^n$ be a λ -eigenvector of A . Write $c = a + ib$ with $a, b \in \mathbb{R}^n$. Then, $Aa + iAb = Ac = \lambda c = \lambda a + i\lambda b$. If $a \neq \underline{0}$ (or respectively, $b \neq \underline{0}$), let $v \in V$ with $[v]_B = a$ (respectively, $[v]_B = b$). Then, v is a λ -eigenvector of α . $\square$

We proceed by induction on $\dim(V)$. The case $\dim(V) = 1$ is trivial. Let $\underline{0} \neq v_1 \in V$ be a λ -eigenvector for α , which exists by Lemma 6.10. Rescaling, we can assume $\|v_1\| = 1$. Then, $\alpha(\langle v_1 \rangle) \subseteq \langle v_1 \rangle$, so by Lemma 6.9, $\alpha(\langle v_1 \rangle^\perp) \subseteq \langle v_1 \rangle^\perp$. Consider $\alpha' = \alpha|_{\langle v_1 \rangle^\perp} : \langle v_1 \rangle^\perp \rightarrow \langle v_1 \rangle^\perp$. Then, for $u, v \in \langle v_1 \rangle^\perp$, we have $\langle \alpha'(u), v \rangle = \langle \alpha(u), v \rangle = \langle u, \alpha(v) \rangle = \langle u, \alpha'(v) \rangle$ so α' is self-adjoint. By induction, $\langle v_1 \rangle^\perp$ has an orthonormal basis B' of eigenvectors for α' . Then, $B = \{v_1\} \cup B'$ is an orthonormal basis of eigenvectors for α . By Lemma 6.8, all the eigenvalues are real. $\square$

Corollary 6.13

- (i) For all symmetric $A \in \mathcal{M}_{n \times n}(\mathbb{R})$, there exists $P \in O(n)$ such that $P^{-1}AP$ is diagonal.
- (ii) For all Hermitian $A \in \mathcal{M}_{n \times n}(\mathbb{C})$, there exists $P \in U(n)$ such that $P^{-1}AP$ is diagonal with real entries.

Proof. Let $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$ and $\langle \cdot, \cdot \rangle$ be the standard dot product on $\mathbb{F}^n$. $A : \mathbb{F}^n \rightarrow \mathbb{F}^n$ is self-adjoint. By the spectral theorem, let $\{v_1, \dots, v_n\} \in \mathbb{F}^n$ be an orthonormal basis of eigenvectors of A with eigenvalues $\lambda_1, \dots, \lambda_n \in \mathbb{R}$. Set $P = (v_1 | \dots | v_n)$, then $\bar{P}^\top P = (\bar{v}_i^\top \cdot v_j)_{ij} = I_n$ so $P \in O(n)$ (or respectively, $U(n)$). Moreover, $AP = (Av_1 | \dots | Av_n) = (\lambda_1 v_1 | \dots | \lambda_n v_n)$ so

$$P^{-1}AP = \bar{P}^\top AP = (\lambda_j \bar{v}_i^\top \cdot v_j)_{ij} = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}.$$

$\square$

Corollary 6.14 – Simultaneous diagonal forms

Let $\mathbb{F} = \mathbb{R}$ (or respectively, $\mathbb{C}$) and V be a finite-dimensional $\mathbb{F}$ -vector space. Let $\varphi, \psi : V \times V \rightarrow \mathbb{F}$ be symmetric bilinear forms (respectively, Hermitian forms). Suppose φ is positive definite, then there is a basis B for V such that both $[\varphi]_{B,B}$ and $[\psi]_{B,B}$ are diagonal and $[\varphi]_{B,B} = I_n$.

Proof. Let $\tilde{B} = \{\tilde{b}_1, \dots, \tilde{b}_n\}$ be an orthonormal basis with respect to φ . Then, $[\varphi]_{\tilde{B},\tilde{B}} = I_n$ and $A = [\psi]_{\tilde{B},\tilde{B}}$ is symmetric (respectively, Hermitian). Let $P \in O(n)$ (respectively, $U(n)$) be such that $D = P\bar{A}^\top P^\top$ is diagonal. Define $b_i \in V$ by $[b_i]_{\tilde{B}} = P^\top [\tilde{b}_i]_{\tilde{B}} = P^\top \mathbf{e}_i$ so that $\{[b_i]_{\tilde{B}} : 1 \leq i \leq n\}$ is an orthonormal basis for $\mathbb{F}^n$. Thus,

$$\varphi(b_i, b_j) = [b_i]_{\tilde{B}}^\top I_n \overline{[b_j]_{\tilde{B}}} = \delta_{ij}.$$

Hence, $B = \{b_1, \dots, b_n\}$ is orthonormal of size $n = \dim(V)$, so it is a basis with $[\varphi]_{B,B} = I_n$. Moreover,

$$\psi(b_i, b_j) = [b_i]_B^\top A \overline{[b_j]_B} = \mathbf{e}_i^\top P A \overline{P}^\top \mathbf{e}_j = \mathbf{e}_i^\top D \mathbf{e}_j = d_{ij}$$

so $[\psi]_{B,B} = D$ is diagonal. □

Let B and C be orthonormal bases for V with respect to inner product φ . Set $P = [\text{id}_V]_B^C$. Then,

$$I_n = [\varphi]_{C,C} = P^\top [\varphi]_{B,B} \overline{P} = P^\top \overline{P} = \overline{P}^\top P = I_n$$

so $P \in O(n)$ if $\mathbb{F} = \mathbb{R}$ and $P \in U(n)$ if $\mathbb{F} = \mathbb{C}$.

Example 6.8

The requirement that φ is positive definite is essential in Corollary 6.14. There is no $P \in \text{GL}_2(\mathbb{R})$ such that $PAP^\top$ and $PBP^\top$ are both diagonal where

$$A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \text{ and } B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Theorem 6.8 – Spectral theorem for unitary maps

Let V be a finite-dimensional complex inner product space and $\alpha \in U(V)$. Then, V has an orthonormal basis of eigenvectors of α with all eigenvalues of modulus one.

Proof. The proof is the same as for self-adjoint maps, but instead of Lemma 6.8, we will use

Lemma 6.11

Let V be an inner product space and $\alpha \in \mathcal{L}(V, V)$ be an isometry.

- (i) All eigenvalues $\lambda \in \mathbb{F}$ of α satisfy $|\lambda| = 1$.
- (ii) Eigenvectors of distinct eigenvalues are orthogonal.

Proof.

- (i) Let $0 \neq v \in V$ be a λ -eigenvector of α . Then,

$$\langle v, v \rangle = \langle \alpha(v), \alpha(v) \rangle = \langle \lambda v, \lambda v \rangle = |\lambda|^2 \langle v, v \rangle.$$

Since $\langle v, v \rangle \neq 0$, we have $|\lambda|^2 = 1$.

- (ii) Suppose $\alpha(v) = \lambda v$ and $\alpha(w) = \mu w$ with $\lambda \neq \mu$. Then,

$$\langle v, w \rangle = \langle \alpha(v), \alpha(w) \rangle = \langle \lambda v, \mu w \rangle = \lambda \overline{\mu} \langle v, w \rangle = \lambda \mu^{-1} \langle v, w \rangle$$

by (i). Since $\lambda \mu^{-1} \neq 1$, we have $\langle v, w \rangle = 0$. □

The rest of the proof is similar. □

This is not true for isometries for real inner product spaces. For example, let

$$A = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \in O(2)$$

with $\theta \in (0, \pi)$. If there exists $P \in O(2)$ such that $PAP^\top$ is diagonal, then A is diagonalisable since $P^\top = P^{-1}$. This is false by Example 5.1.

Definition 6.28 – Normal map

Let V be a finite-dimensional complex inner product space. $\alpha \in \mathcal{L}(V, V)$ is normal if $\alpha\alpha^* = \alpha^*\alpha$.

Self-adjoint maps and unitary maps are normal. On example sheet 4, we will prove that if α is normal, then V has an orthonormal basis of eigenvectors of α .

Since diagonal matrices commute, this is a "if and only if" statement. Also,

$$\underbrace{\begin{pmatrix} r_1 e^{i\theta_1} & & 0 \\ & \ddots & \\ 0 & & r_n e^{i\theta_n} \end{pmatrix}}_{\text{normal}} = \underbrace{\begin{pmatrix} r_1 & & 0 \\ & \ddots & \\ 0 & & r_n \end{pmatrix}}_{\text{self-adjoint}} \underbrace{\begin{pmatrix} e^{i\theta_1} & & 0 \\ & \ddots & \\ 0 & & e^{i\theta_n} \end{pmatrix}}_{\text{unitary}}.$$